

Laboratory Evaluation of Gut Dysfunction



DAN LUKACZER, ND

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Dan Lukaczer, ND, disclosed no relevant financial relationships with any ineligible companies.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Align the analytes of laboratory stool tests with their associated gut DIGIN dysfunction.
2. Cite common clinical scenarios that warrant stool testing.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

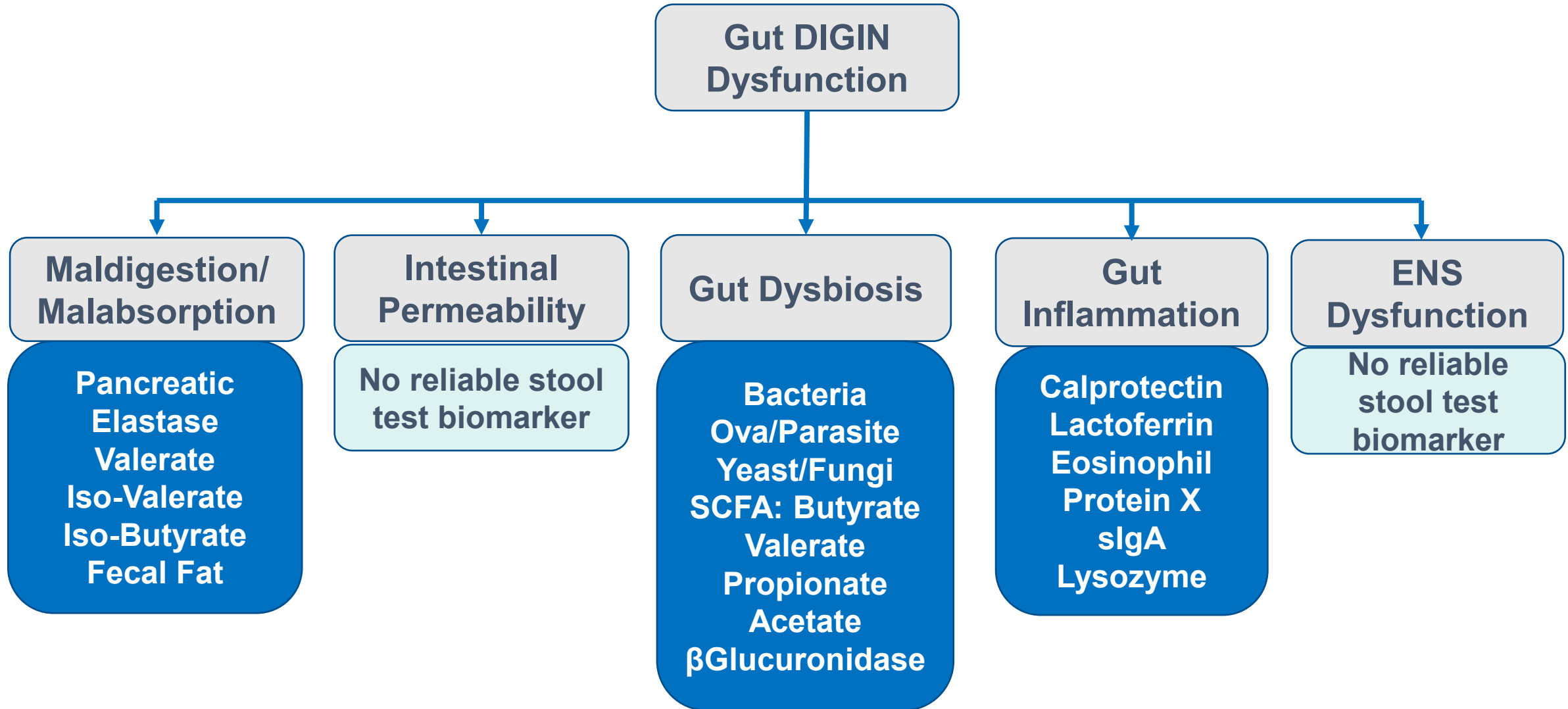
1. Discern and identify clinical scenarios of gut dysfunction when laboratory stool testing is warranted.
2. Interpret laboratory stool test biomarkers of gut dysfunction.

Clinical Focus

- Alignment of stool test analytes with gut DIGIN dysfunctions.
- Other gut DIGIN dysfunction diagnostic modalities outside the scope of this training.



DIGIN Dysfunctions and Stool Test Biomarkers



Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

This graphic categorizes stool test biomarkers associated with key DIGIN gut dysfunctions.

- **Maldigestion/Malabsorption**

Associated stool biomarkers:

- Pancreatic elastase
- Valerate
- Iso-valerate
- Iso-butyrate
- Fecal fat

- **Intestinal Permeability**

There is **no reliable stool test biomarker** currently available for this dysfunction.

- **Gut Dysbiosis**

Associated stool biomarkers:

- Bacteria
- Ova/parasites
- Yeast/fungi
- Short-chain fatty acids (S C F A): butyrate, valerate, propionate, acetate
- Beta-glucuronidase

- **Gut Inflammation**

Associated stool biomarkers:

- Calprotectin
- Lactoferrin
- Eosinophil protein X
- Secretory I g A (s I g A)
- Lysozyme

- **Enteric Nervous System (ENS) Dysfunction**

There is **no reliable stool test biomarker** currently available for this dysfunction.

Stool Test Biomarkers/Lab Analytes

GI	Associated Biomarkers
Digestion/Absorption	• Pancreatic elastase • Valerate • Isovalerate • Iso-butyrate • Fecal fat
Inflammation	• Calprotectin • Lactoferrin • Eosinophil protein X • Secretory IgA (sIgA) • Lysozyme

Stool Test Biomarkers/Lab Analytes

GI	Associated Biomarkers
Microbiome: Organisms	• Ova and parasite (protozoa & helminths) • Bacteria • Viruses • Fungal, yeast, candida
Microbiome: Metabolites	• Short-chain fatty acids (SCFAs) • Butyrate • Acetate • Propionate • Valerate • Beta glucuronidase • Fecal pH

References

BIOMARKERS/LAB ANALYTES

Pancreatic Elastase:

- Othman MO, Harb D, Barkin JA. Introduction and practical approach to exocrine pancreatic insufficiency for the practicing clinician. *Int J Clin Pract*. 2018;72(2):e13066. doi:10.1111/ijcp.13066
- Lam KW, Leeds J. How to manage: patient with a low faecal elastase. *Frontline Gastroenterol*. 2019;12(1):67-73. doi:10.1136/flgastro-2018-101171
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- El Kurdi B, Babar S, El Iskandarani M, et al. Factors that affect prevalence of small intestinal bacterial overgrowth in chronic pancreatitis: a systematic review, meta-analysis, and meta-regression. *Clin Transl Gastroenterol*. 2019;10(9):e00072. doi:10.14309/ctg.0000000000000072
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Products of Protein Breakdown (Valerate, Iso-Valerate, Iso-Butyrate):

- Zarling EJ, Ruchim MA. Protein origin of the volatile fatty acids isobutyrate and isovalerate in human stool. *J Lab Clin Med*. 1987;109(5):566-570.
- Rasmussen HS, Holtug K, Mortensen PB. Degradation of amino acids to short-chain fatty acids in humans. An in vitro study. *Scand J Gastroenterol*. 1988;23(2):178-182. doi:10.3109/00365528809103964
- Nakamura T, Tabeke K, Terada A, et al. Short-chain carboxylic acid in the feces in patients with pancreatic insufficiency. *Acta Gastroenterol Belg*. 1993;56(5-6):326-331.
- Yao CK, Muir JG, Gibson PR. Review article: insights into colonic protein fermentation, its modulation and potential health implications. *Aliment Pharmacol Ther*. 2016;43(2):181-196. doi:10.1111/apt.13456

Fecal Fat:

- Pongprasobchai S. Maldigestion from pancreatic exocrine insufficiency. *J Gastroenterol Hepatol*. 2013;28(Suppl 4):99-102. doi:10.1111/jgh.12406
- Corinaldesi R, Stanghellini V, Barbara G, Tomassetti P, De Giorgio R. Clinical approach to diarrhea. *Intern Emerg Med*. 2012;7(Suppl 3):S255-S262. doi:10.1007/s11739-012-0827-4

References

BIOMARKERS/LAB ANALYTES

Eosinophil Protein X:

- Abedin N, Seemann T, Kleinfeld S, et al. Fecal eosinophil cationic protein is a diagnostic and predictive biomarker in young adults with inflammatory bowel disease. *J Clin Med*. 2019;8(12):2025. doi:10.3390/jcm8122025

Secretory IgA:

- Hansen IS, Baeten DLP, den Dunnen J. The inflammatory function of human IgA. *Cell Mol Life Sci*. 2019;76(6):1041-1055. doi:10.1007/s00018-018-2976-8

Short-Chain Fatty Acids (Butyrate, Acetate, Propionate, Valerate):

- Silva YP, Bernardi A, Frozza RL. The role of short-chain fatty acids from gut microbiota in gut-brain communication. *Front Endocrinol (Lausanne)*. 2020;11:25. doi:10.3389/fendo.2020.00025
- Nogal A, Valdes AM, Menni C. The role of short-chain fatty acids in the interplay between gut microbiota and diet in cardio-metabolic health. *Gut Microbes*. 2021;13(1):1-24. doi:10.1080/19490976.2021.1897212
- Iwasaki M, Akiba Y, Kaunitz JD. Duodenal chemosensing of short-chain fatty acids: implications for GI diseases. *Curr Gastroenterol Rep*. 2019;21(8):35. doi:10.1007/s11894-019-0702-9
- Høverstad T, Bjørneklett A, Fausa O, Midtvedt T. Short-chain fatty acids in the small-bowel bacterial overgrowth syndrome. *Scand J Gastroenterol*. 1985;20(4):492-499. doi:10.3109/00365528509089686

Beta-Glucuronidase:

- Flores R, Shi J, Gail MH, Gajer P, Ravel J, Goedert JJ. Association of fecal microbial diversity and taxonomy with selected enzymatic functions. *PLoS One*. 2012;7(6):e39745. doi:10.1371/journal.pone.0039745

Secondary Bile Acids:

- Zeng H, Umar S, Rust B, Lazarova D, Bordonaro M. Secondary bile acids and short chain fatty acids in the colon: a focus on colonic microbiome, cell proliferation, inflammation, and cancer. *Int J Mol Sci*. 2019;20(5):1214. doi:10.3390/ijms20051214

References

BIOMARKERS/LAB ANALYTES

Fecal pH:

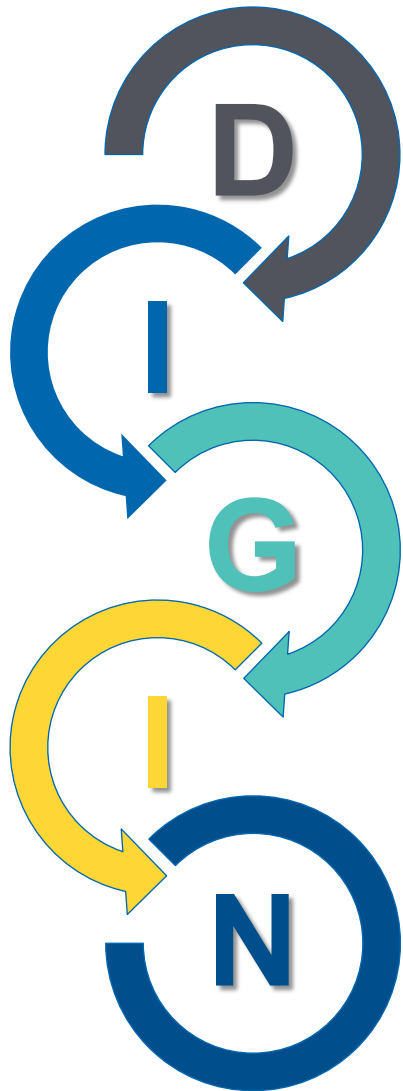
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- Animalu CN, Wallace MR, Syed SZ, et al. Bacterial overgrowth syndrome workup. Medscape. Updated September 20, 2024. Accessed June 4, 2025. <https://emedicine.medscape.com/article/212861-workup>
- Omer A, Quigley EMM. Carbohydrate maldigestion and malabsorption. *Clin Gastroenterol Hepatol*. 2018;16(8):1197-1199. doi:10.1016/j.cgh.2018.01.048
- Osuka A, Shimizu K, Ogura H, et al. Prognostic impact of fecal pH in critically ill patients. *Crit Care*. 2012;16(4):R119. doi:10.1186/cc11413
- Vahouny GV, Kritchevsky D, eds. *Dietary Fiber in Health and Disease*. Springer Science & Business Media; 2013.
- Russell WR, Gratz SW, Duncan SH, et al. High-protein, reduced-carbohydrate weight-loss diets promote metabolite profiles likely to be detrimental to colonic health. *Am J Clin Nutr*. 2011;93(5):1062-1072.
- Ohigashi S, Sudo K, Kobayashi D, et al. Changes of the intestinal microbiota, short chain fatty acids, and fecal pH in patients with colorectal cancer. *Dig Dis Sci*. 2013;58(6):1717-1726. doi:10.1007/s10620-012-2526-4
- Down PF, Agostini L, Murison J, Wrong OM. The interrelations of faecal ammonia, pH and bicarbonate: evidence of colonic absorption of ammonia. *Clin Sci*. 1972;43(1):101-114. doi:10.1042/cs0430101

Clinical Focus

- Stool test biomarkers of maldigestion and malabsorption.



Interpretation of Analytes of Maldigestion and Malabsorption Dysregulation



Digestion/Absorption

Intestinal Permeability

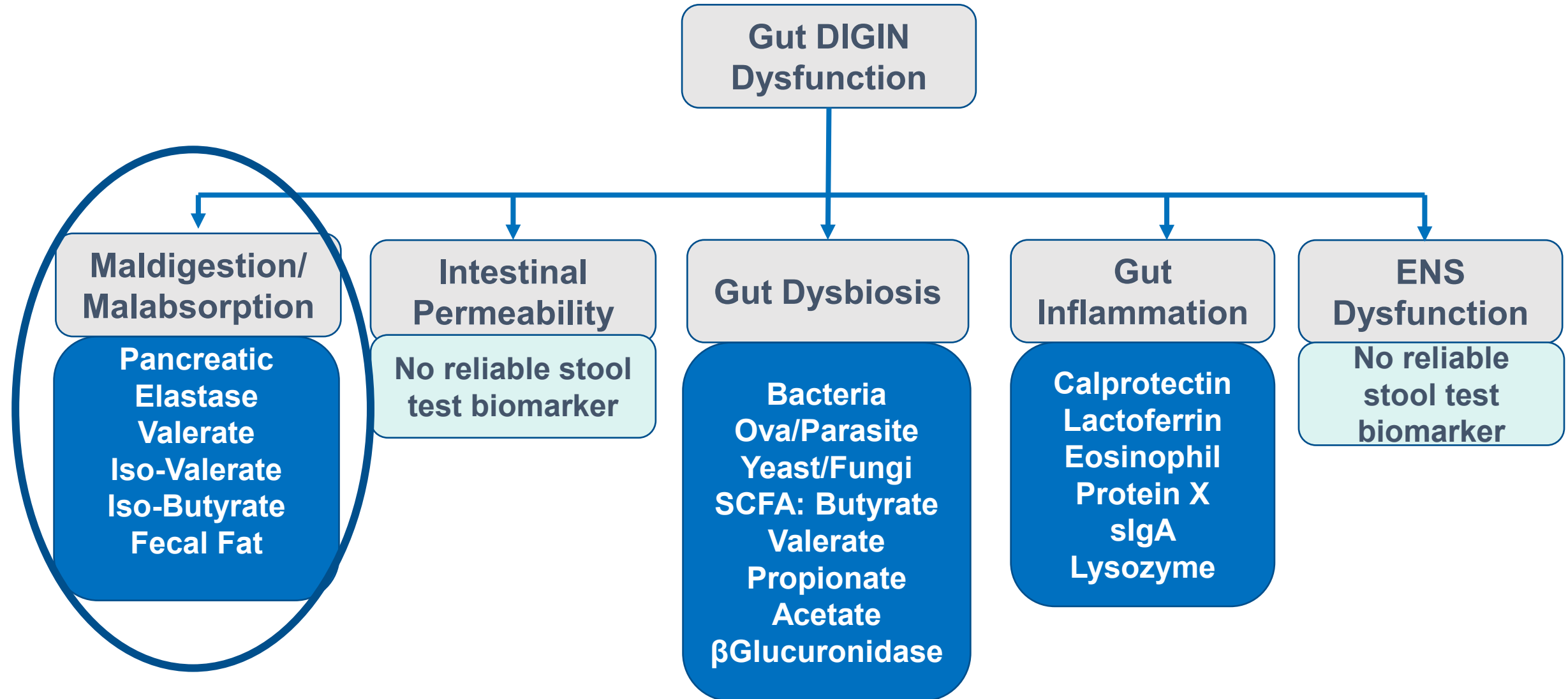
Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System (Enteric)

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

DIGIN Dysfunctions and Stool Test Biomarkers



Pancreatic Elastase

- Proteolytic enzyme secreted exclusively by the human pancreas
- Reflects overall enzyme production
 - Amylase, lipase, and protease
- Not affected by supplemental enzymes
- Non-invasive marker for evaluating exocrine pancreatic function
 - Sensitivity = 90-100%
 - Specificity = 93-98%

Reference

PANCREATIC ELASTASE

- Stein J, Jung M, Sziegoleit A, Zeuzem S, Caspary WF, Lembcke B. Immunoreactive elastase I: clinical evaluation of a new noninvasive test of pancreatic function. *Clin Chem*. 1996;42(2):222-226.
- Löser C, Möllgaard A, Fölsch UR. Faecal elastase 1: a novel, highly sensitive, and specific tubeless pancreatic function test. *Gut*. 1996;39(4):580-586. doi:10.1136/gut.39.4.580
- Mattar R, Lima GA, da Costa MZ, Silva-ETTO JM, Guarita D, Carrilho FJ. Comparison of fecal elastase 1 for exocrine pancreatic insufficiency evaluation between ex-alcoholics and chronic pancreatitis patients. *Arq Gastroenterol*. 2014;51(4):297-301. doi:10.1590/S0004-28032014000400006
- Vujasinovic M, Tepes B, Makuc J, et al. Pancreatic exocrine insufficiency, diabetes mellitus and serum nutritional markers after acute pancreatitis. *World J Gastroenterol*. 2014;20(48):18432-18438. doi:10.3748/wjg.v20.i48.18432
- Lindkvist B. Diagnosis and treatment of pancreatic exocrine insufficiency. *World J Gastroenterol*. 2013;19(42):7258-7266. doi:10.3748/wjg.v19.i42.7258

Pancreatic Elastase: Clinical Context

- Can be used for initial determination of pancreatic insufficiency and to monitor function in patients under treatment
- Patients in whom testing may be useful include:
 - **Unexplained diarrhea**
 - **Weight loss**
 - **Other signs of malabsorption/nutrient deficiencies**
 - **Abdominal pain**
- Exocrine pancreatic insufficiency may occur secondary to:
 - **Chronic pancreatitis, diabetes, celiac disease, inflammatory bowel disease, cystic fibrosis, alcohol consumption, gallstone disease**

Decreased Pancreatic Elastase: Clinical Context

Root Causes:

- Chronic Pancreatitis
- Cystic Fibrosis
- Pancreatic Cancer
- Type 1 and Type 2 Diabetes
- Celiac Disease Zollinger-Ellison Syndrome
- Aging
- Other Genetic Disorders

Functional Medicine Connections:

- Chronic Stress and HPA-Axis Dysfunction
- Small Intestinal Bacterial Overgrowth
- Food Sensitivities
- Low Stomach Acid (Hypochlorhydria)
- Neurodegenerative or Neurological Conditions
- Dysbiosis and Gut-Immune Interactions

References

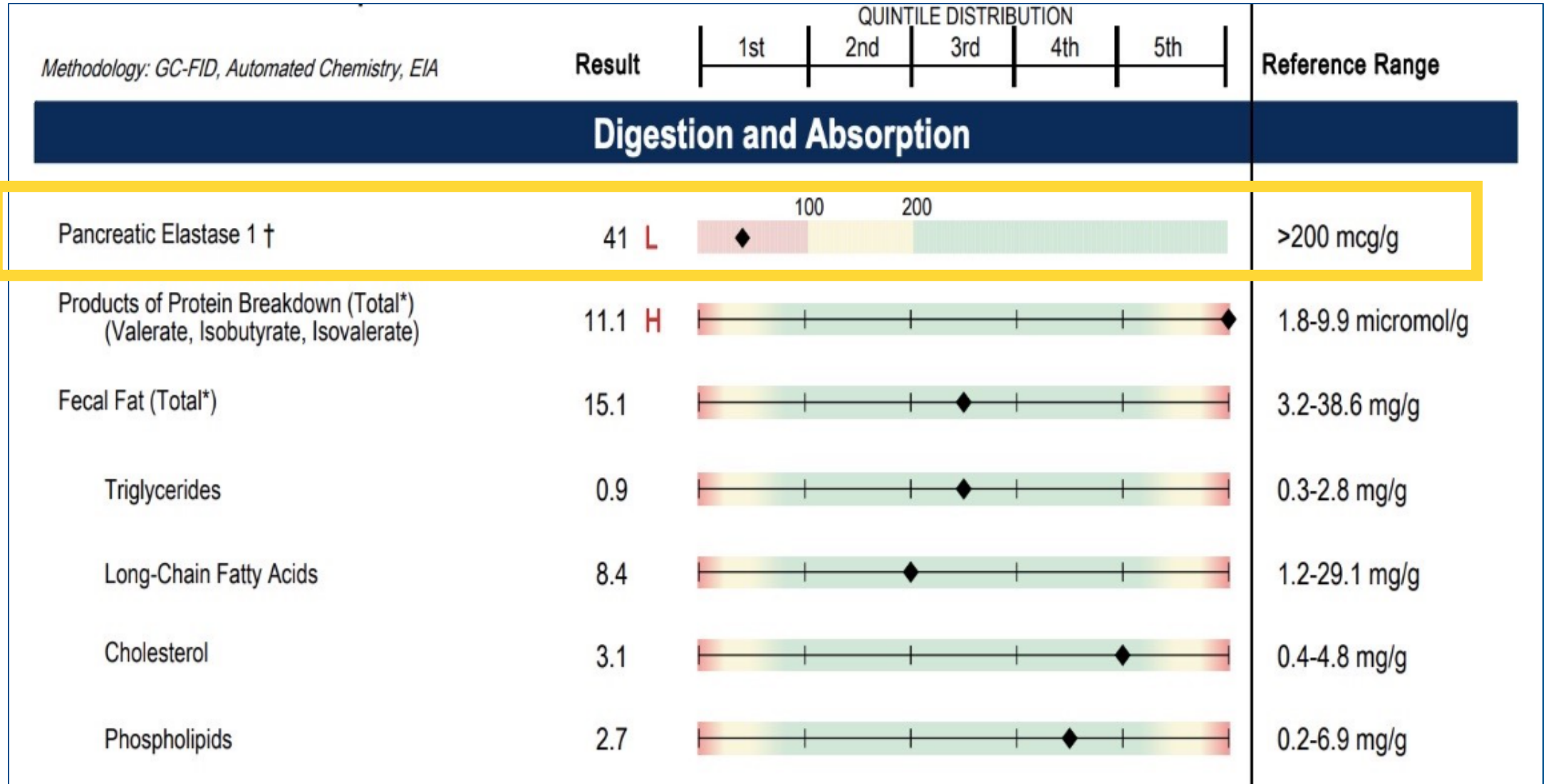
PANCREATIC ELASTASE: CLINICAL CONTEXT

- Löhr JM, Dominguez-Munoz E, Rosendahl J, et al. United European Gastroenterology evidence-based guidelines for the diagnosis and therapy of chronic pancreatitis (HaPanEU). *United European Gastroenterol J*. 2017;5(2):153-199. doi:10.1177/2050640616684695
- Di Martino M, de la Hoz Rodriguez Á, Saibanti A, et al. Pancreatic exocrine insufficiency after pancreatic resection: a systematic review. *BMC Surg*. 2025;25(1):53. doi:10.1186/s12893-025-02787-y
- Dumonceau JM, Macias-Gomez C. Endoscopic management of complications of chronic pancreatitis. *World J Gastroenterol*. 2013;19(42):7308-7315. doi:10.3748/wjg.v19.i42.7308

Pancreatic Elastase: Interpretation

>350 µg/g	Normal pancreatic function
200-350 µg/g	Declining pancreatic function Consider supplementation
100-200 µg/g	Moderate pancreatic insufficiency Supplement with broad array of pancreatic enzymes
<100 µg/g	Severe pancreatic insufficiency Supplement with broad array of pancreatic enzymes

Pancreatic Elastase: Interpretation



Representative Example: Stool Test Biomarkers

DIGESTION	Result	Reference
Steatocrit	<dl 	< 15 %
Elastase-1	>750 	> 200 ug/g

DIGESTION	Result	Reference
Steatocrit	<dl 	< 15 %
Elastase-1	356 	> 200 ug/g

Image courtesy of Dan Lukaczer,, ND

Products of Protein Breakdown

- There are three products of protein breakdown: **valerate**, **isovalerate**, and **isobutyrate**.
- These breakdown products are the result of the **anaerobic fermentation of polypeptides and amino acids by gut flora**.
- Interpretation: impaired protein digestion.

Valerate

Isovalerate

Isobutyrate

Products of Protein Breakdown: Clinical Context

ROOT CAUSES:

- Hypochlorhydria resulting in poor protein digestion
- Low secretion of protein-digesting enzymes by the pancreas
- Poor absorption of protein due to inflammation/damage to the gut lining (i.e., celiac, Crohn's disease)
- Dysbiosis: small intestinal bacterial overgrowth (SIBO)
- Hypochlorhydria and SIBO are not well-evaluated by stool testing

References

PRODUCTS OF PROTEIN BREAKDOWN

- **Exocrine Pancreatic Insufficiency:** Pezzilli R, Andriulli A, Bassi C, et al. Exocrine pancreatic insufficiency in adults: a shared position statement of the Italian Association for the Study of the Pancreas. *World J Gastroenterol*. 2013;19(44):7930-7946. doi:10.3748/wjg.v19.i44.7930.
- **SIBO:** Bures J, Cyrany J, Kohoutova D, et al. Small intestinal bacterial overgrowth syndrome. *World J Gastroenterol*. 2010;16(24):2978-2990. doi:10.3748/wjg.v16.i24.2978
- **Hypochlorhydria:** Revaiah PC, Kochhar R, Rana SV, et al. Risk of small intestinal bacterial overgrowth in patients receiving proton pump inhibitors versus proton pump inhibitors plus prokinetics. *JGH Open*. 2018;2(2):47-53. doi:10.1002/jgh3.12045
- **Increased Protein Consumption:** Geypens B, Claus D, Evenepoel P, et al. Influence of dietary protein supplements on the formation of bacterial metabolites in the colon. *Gut*. 1997;41(1):70-76. doi:10.1136/gut.41.1.70

Products of Protein Breakdown

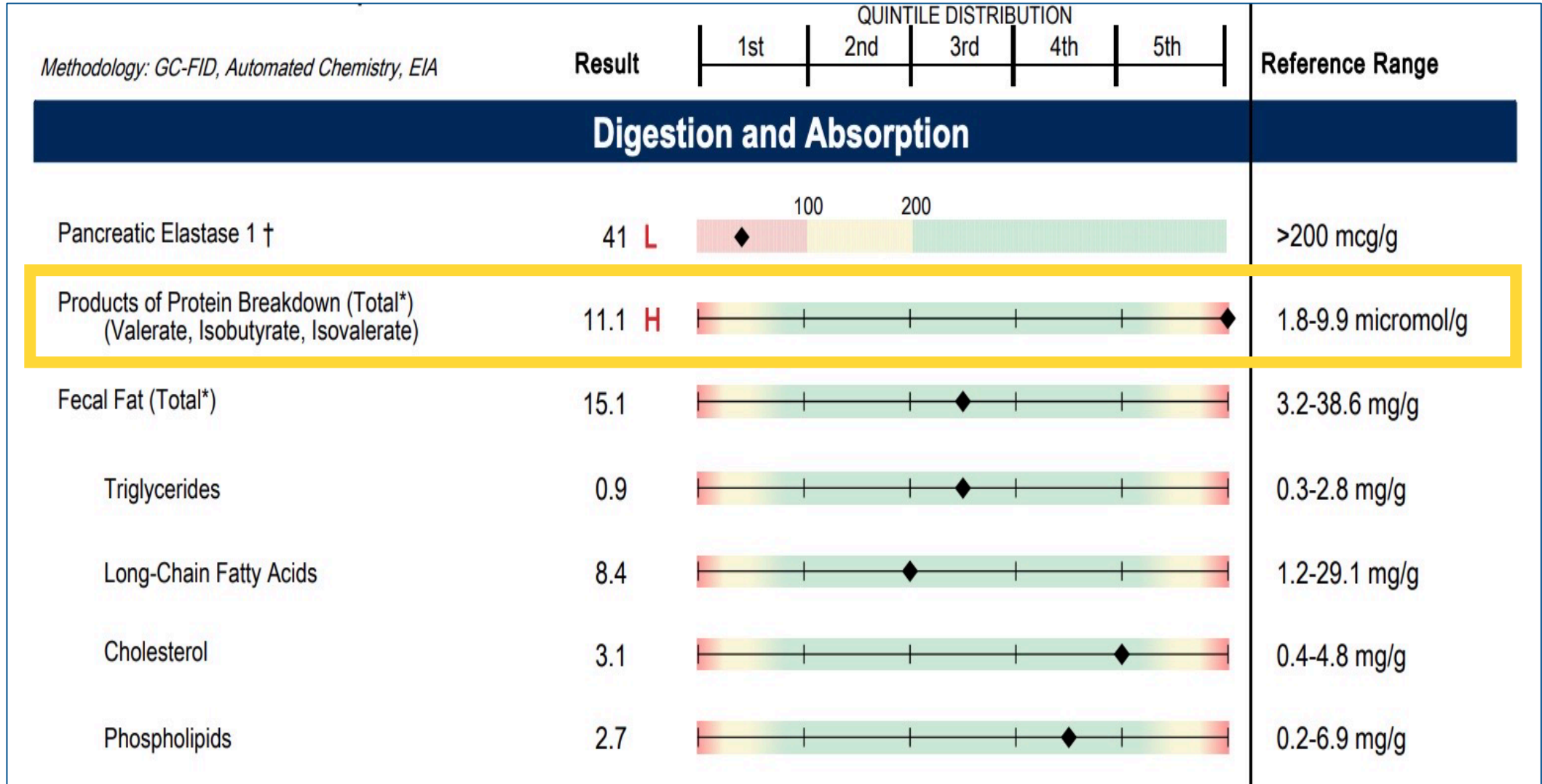


Image courtesy of Elizabeth Boham, MD, MS, RD

Fecal Fat Testing

Also known as: fecal fat stain, quantitative stool fat

- Measures the number of fat globules in a stool sample
- Used to identify patients with steatorrhea/fat malabsorption
- Common causes for an elevated fecal fat level include: **celiac disease, exocrine pancreas insufficiency, Crohn's disease, enteritis, or liver/gallbladder disease**

Fecal Fat Testing: Clinical Context

ROOT CAUSES:

- Steatorrhea: Condition characterized by excessive fat in stool, a hallmark of maldigestion
- Pancreatic and Gallbladder Conditions
 - Bile acid insufficiency
 - Bile acid malabsorption
 - Pancreatic insufficiency

Fecal Fat Testing

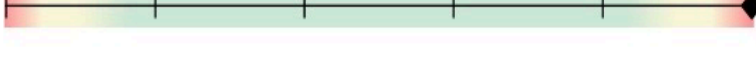
Digestion and Absorption				
Pancreatic Elastase 1 †	>500		>200 mcg/g	
Products of Protein Breakdown (Total*) (Valerate, Isobutyrate, Isovalerate)	4.9		1.8-9.9 micromol/g	
Fecal Fat (Total*)	48.9 H		3.2-38.6 mg/g	
Triglycerides	<DL L		0.3-2.8 mg/g	
Long-Chain Fatty Acids	33.9 H		1.2-29.1 mg/g	
Cholesterol	6.8 H		0.4-4.8 mg/g	
Phospholipids	8.2 H		0.2-6.9 mg/g	

Image courtesy of Elizabeth Boham, MD, MS, RD

Fecal Fat Testing

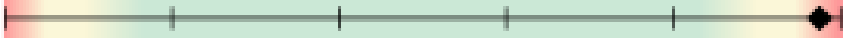
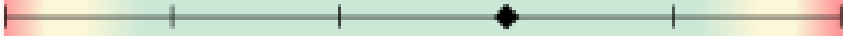
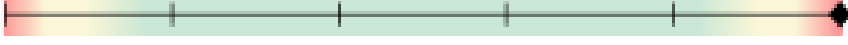
Digestion and Absorption				
Pancreatic Elastase 1 †	>500		>200 mcg/g	
Products of Protein Breakdown (Total*) (Valerate, Isobutyrate, Isovalerate)	7.2		1.8-9.9 micromol/g	
Fecal Fat (Total*)	>54.4 H		3.2-38.6 mg/g	
Triglycerides	3.3 H		0.3-2.8 mg/g	
Long-Chain Fatty Acids	28.5		1.2-29.1 mg/g	
Cholesterol	2.1		0.4-4.8 mg/g	
Phospholipids	>20.5 H		0.2-6.9 mg/g	

Image courtesy of Dan Lukaczer, ND

Laboratory Evaluation of Gut Dysfunction: Part 2

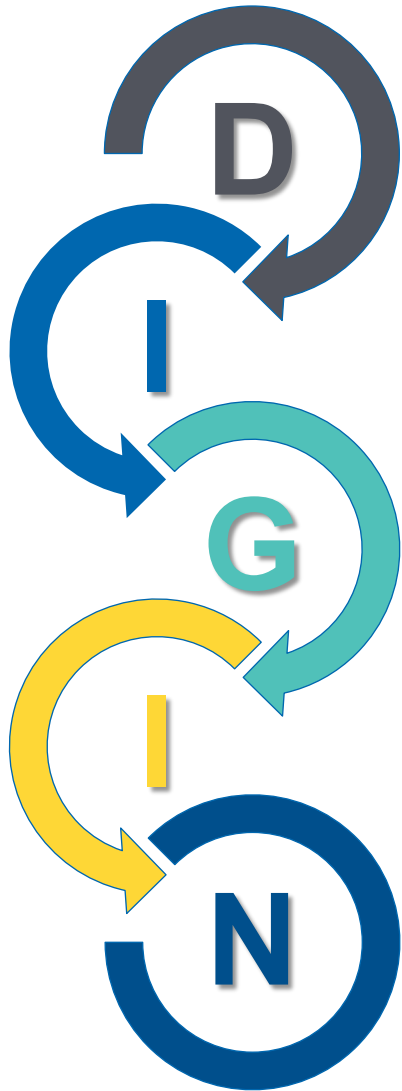
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Stool test biomarkers of gut inflammation and immune dysregulation.



Interpretation of Analytes of Inflammation and Immune Dysregulation



Digestion/Absorption

Intestinal Permeability

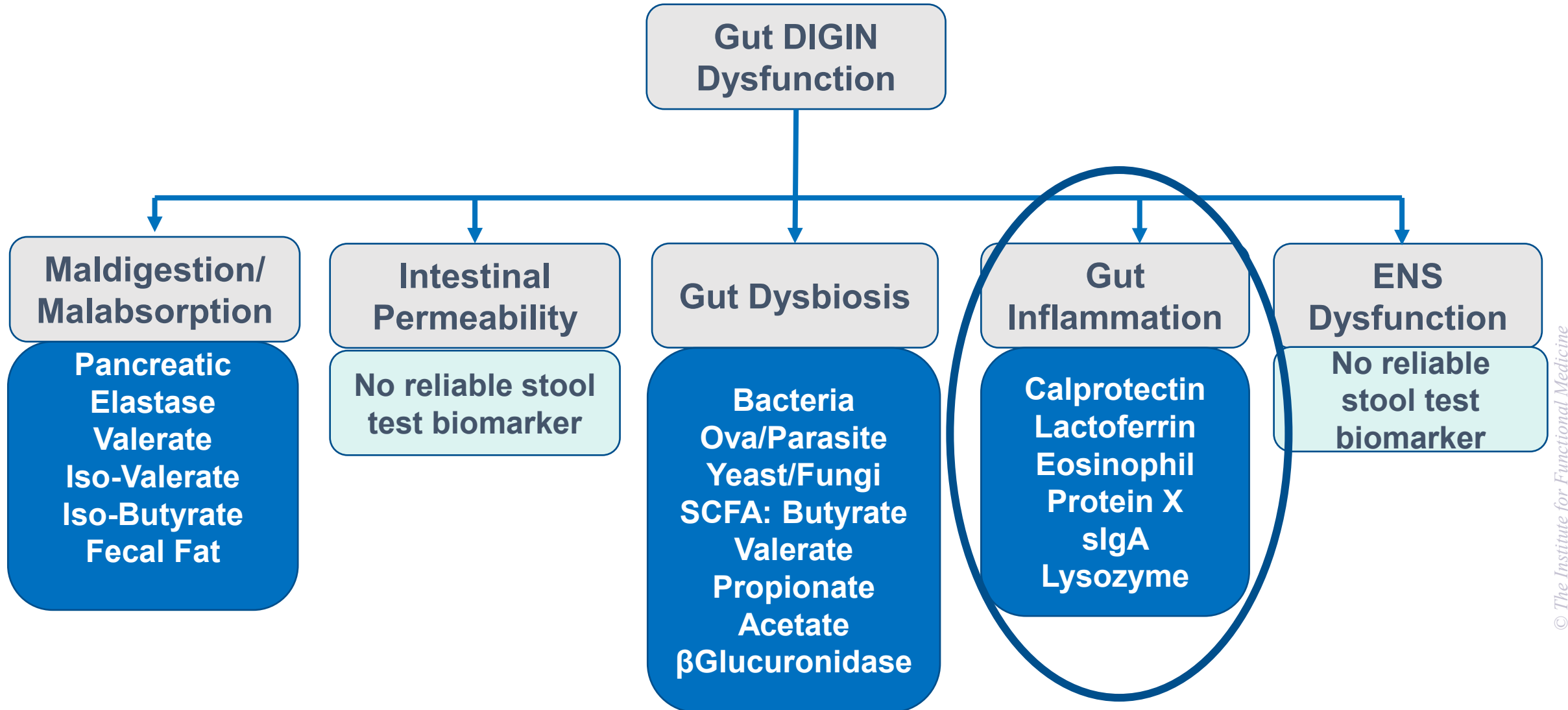
Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System (Enteric)

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

DIGIN Dysfunctions and Stool Test Biomarkers



Stool Test Biomarkers: Inflammation/Immunity Markers

1) Local inflammatory metabolites

Calprotectin

Lactoferrin

2) Immune markers

Eosinophil protein X

Secretory IgA

Lysozyme

Local Inflammatory Markers: Clinical Context

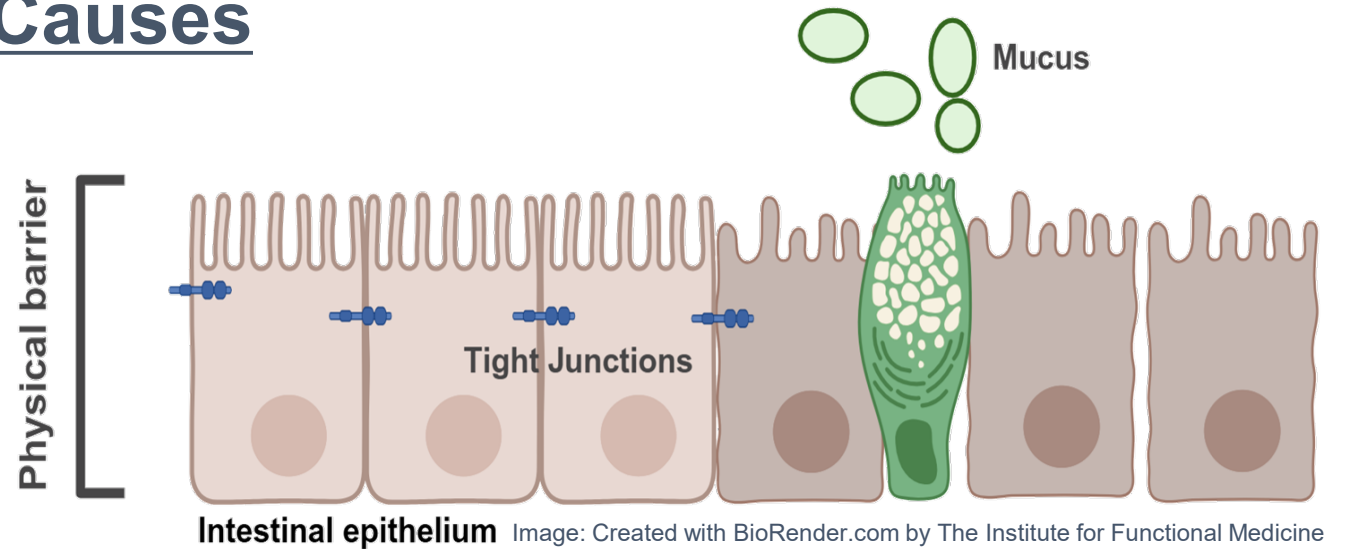
Calprotectin and Lactoferrin

Calprotectin: A calcium and zinc-binding protein found in neutrophils

Lactoferrin: Iron-binding glycoprotein released by neutrophils

Clinical Context and Root Causes

- Elevated in the stool when there is intestinal inflammation
- Regulates the physical barriers of gut immunity



Long Description: Local Inflammatory Markers

A diagram illustrating the **physical barrier of the intestinal lining**.

- The **intestinal epithelium** is composed of a single layer of tightly packed epithelial cells.
- These cells are connected by **tight junctions**, which are specialized protein structures that regulate what passes between the cells, maintaining barrier integrity.
- One of the cells is a **goblet cell**, shown in green, which secretes **mucus** onto the surface of the epithelium.
- The **mucus layer** traps pathogens and particles, providing an additional protective barrier above the cells.
- Together, the mucus layer, epithelial cells, and tight junctions make up the **intestinal physical barrier**, helping prevent harmful substances from entering the body while allowing nutrient absorption.

Calprotectin

- Found in extra lysosomal cytosol of neutrophils
- Inhibitory effect on microbial zinc dependent enzymes
- Result = bacteriostatic activity

Calprotectin

Elevated in:

- Inflammatory bowel disease
- Post-infectious irritable bowel syndrome
- Gastrointestinal cancers
- Certain gastrointestinal infections
- NSAID enteropathy
- Food allergy
- Chronic pancreatitis

Use Calprotectin to Differentiate IBD vs IBS

Calprotectin is an FDA-cleared biomarker that is highly accurate and capable of differentiating IBS from IBD.

A person with positive Rome criteria for IBS and a normal calprotectin (<50 µg/g) has virtually
NO CHANCE OF HAVING IBD!

- Banerjee A, Srinivas M, Eyre R, et al. Faecal calprotectin for differentiating between irritable bowel syndrome and inflammatory bowel disease: a useful screen in daily gastroenterology practice. *Frontline Gastroenterol.* 2015;6(1):20-26. doi:10.1136/flgastro-2013-100429
- Shi JT, Zhang Y, She Y, Goyal H, Wu ZQ, Xu HG. Diagnostic utility of non-invasive tests for inflammatory bowel disease: an umbrella review. *Front Med (Lausanne).* 2022;9:920732. doi:10.3389/fmed.2022.920732
- Petryszyn P, Staniak A, Wolosińska A, Ekk-Cierniakowski P. Faecal calprotectin as a diagnostic marker of inflammatory bowel disease in patients with gastrointestinal symptoms: meta-analysis. *Eur J Gastroenterol Hepatol.* 2019;31(11):1306-1312. doi:10.1097/MEG.0000000000001509

Calprotectin

A meta-analysis published in 2019 provides a useful calculation of potential interest to payors, as well as to clinicians and patients.

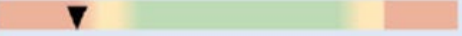
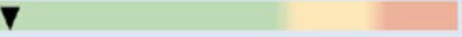
Evaluated 19 studies and found that in adults being evaluated for IBD, ***screening by measuring calprotectin levels would produce a 67% reduction in the number of adults undergoing colonoscopy.***

Calprotectin: Interpretation

< 50 µg/g	No significant inflammation
50-120 µg/g	Indicates some GI inflammation: IBD, infection, polyps, neoplasia, NSAIDS
> 120 µg/g	Significant inflammation; referral may be indicated to determine pathology
> 250 µg/g	Active disease present; predicts imminent relapse in treated patients

Lab Examples of Gut Inflammation

IMMUNE RESPONSE

Secretory IgA	500 L		510 - 2010 ug/g
Anti-gliadin IgA	104		< 175 U/L
Eosinophil Activation Protein (EDN, EPX)	0.09		< 2.34 ug/g

INFLAMMATION

Calprotectin	115		< 173 ug/g
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Image courtesy of Elizabeth McMillan, CNS, LDN

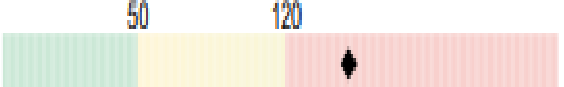
Inflammation and Immunology				
Calprotectin †	148 H		50 120	<=50 mcg/g
Eosinophil Protein X (EPX) †	0.3		0.5 2.7	<=2.7 mcg/g
Fecal secretory IgA	430		680 2040	<=2,040 mcg/mL

Image courtesy of Dan Lukaczer, ND

Additional Lab Examples of Gut Inflammation

Lactoferrin

	Result	Unit	L	WRI	H	Reference Interval
Lactoferrin	17.6	µg/mL				<7.3

Information

- **Lactoferrin** is a quantitative GI specific marker of inflammation used to diagnose and differentiate IBD from IBS and to monitor patient inflammation levels during active and remission phases of inflammatory bowel disease.

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Calprotectin vs Lactoferrin Summary

- Similar ability predicting clinical relapse of IBD.
- Fairly similar ability differentiating IBD from IBS.
- Both are sensitive markers of inflammation in the gut.

Stool Test Biomarkers: Inflammation/Immunity Markers

1) Local inflammatory metabolites

Calprotectin

Lactoferrin

2) Immune markers

Eosinophil Protein X

Secretory IgA

Lysozyme

Immune Markers: Clinical Context

Eosinophil Protein X, Secretory IgA, Lysozyme

Eosinophil protein X: released by activated eosinophils

Secretory IgA: main immunoglobulin in mucosal secretions

Lysozyme: antimicrobial enzyme secreted by neutrophils and epithelial cells

Clinical Context and Root Causes:

- Released if mucosal immune system is activated
 - Food allergies/sensitivities
 - Eosinophilic gastritis
 - Parasitic infections
 - Dysbiosis
 - Environmental toxins
 - Autoimmune diseases

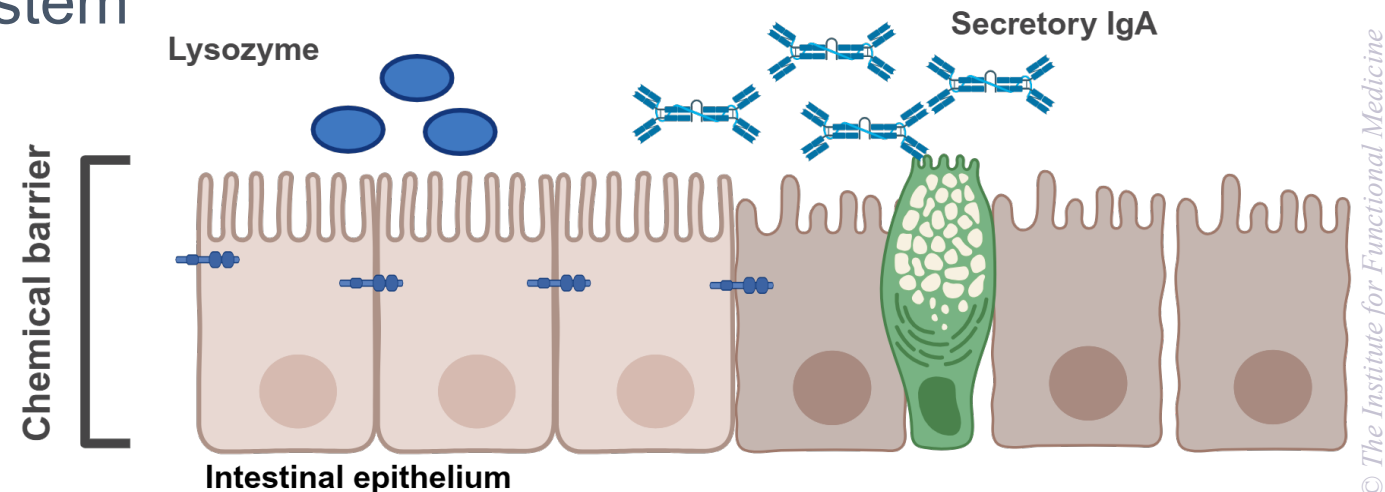


Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Immune Markers

This diagram illustrates the **chemical barrier** of the intestinal lining, which complements the physical barrier in defending against pathogens.

- The image shows intestinal epithelial cells joined by tight junctions.
- A green goblet cell is depicted, which secretes mucus.
- Above the epithelial layer, two key immune-related chemical defenses are labeled:
 - **Lysozyme** (shown as blue ovals), an enzyme that breaks down bacterial cell walls.
 - **Secretory I g A (s I g A)** (depicted as Y-shaped blue molecules), an antibody that binds to pathogens and prevents them from attaching to and penetrating the epithelial lining.
- Together, **lysozyme and secretory I g A** contribute to the **chemical barrier** by neutralizing harmful microbes and maintaining immune surveillance in the gut mucosa.

Eosinophilic Protein X

- Released in **eosinophil** degranulation
- Sensitive marker of GI **inflammation**
- May predict **relapse in IBD**

May be elevated with:

- Inflammatory bowel disease
- Celiac disease
- Parasites
- Allergic reaction
- Less common:
 - GERD
 - Chronic diarrhea
 - Chronic alcoholism
 - Protein-losing enteropathy

Reference

EOSINOPHILIC PROTEIN X

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Eosinophilic Protein X

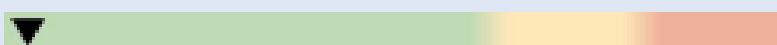
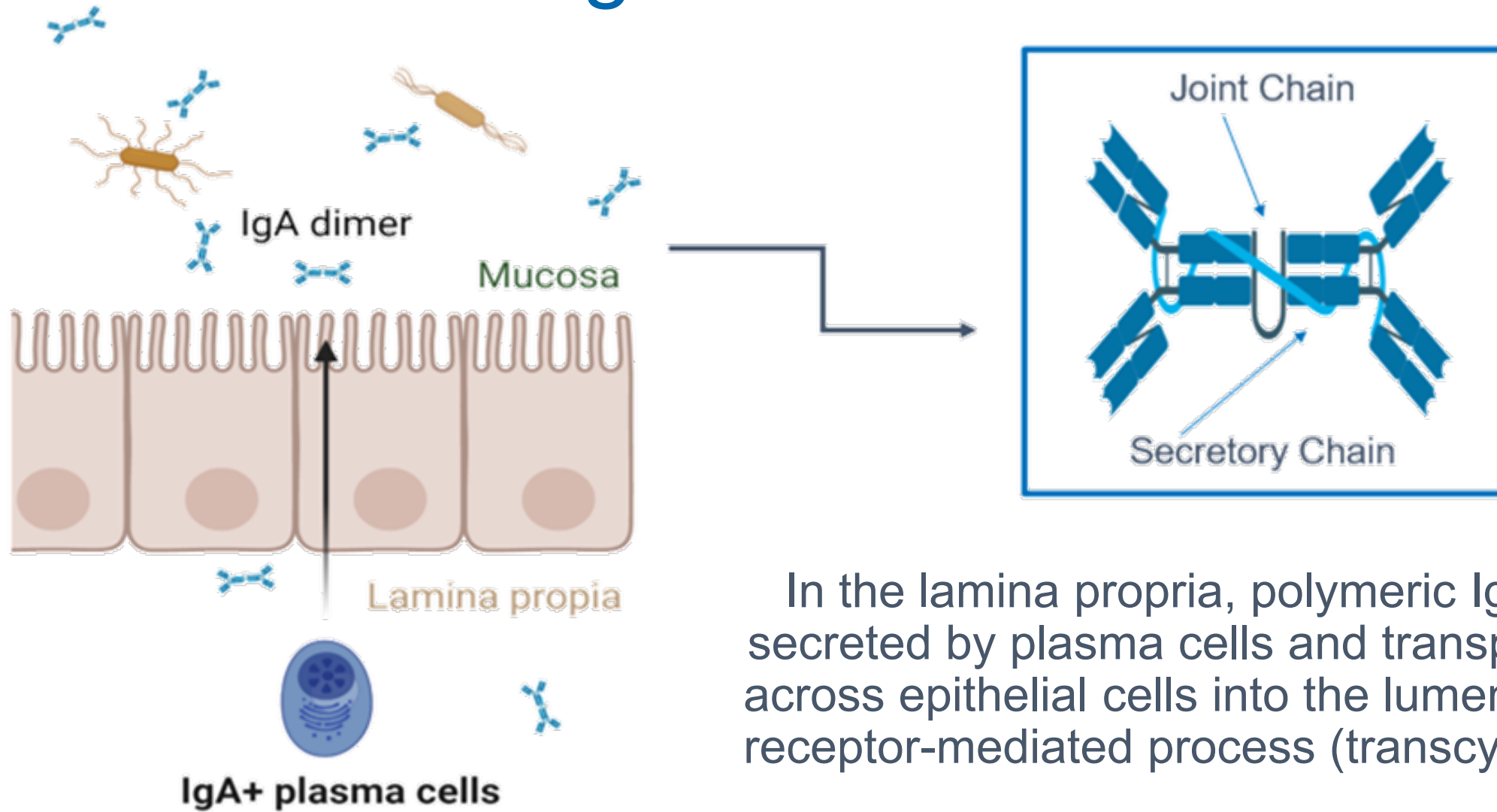
IMMUNE RESPONSE				
Secretory IgA	2820 H			510 - 2010 ug/g
Anti-qliadin IgA	63			< 175 U/L
Eosinophil Activation Protein (EDN, EPX)	0.73			< 2.34 ug/g
INFLAMMATION				
Calprotectin	14			< 173 ug/g

Image Courtesy of Dan Lukaczer, ND

Secretory IgA (sIgA)

- Predominant immunoglobulin released onto the surface of the GI mucosa.
- Binds to and neutralizes microbes and other antigens before they can cross the mucosal barrier.

IgA Secretion



In the lamina propria, polymeric IgA is secreted by plasma cells and transported across epithelial cells into the lumen via a receptor-mediated process (transcytosis).

- Image: Adapted from "IgA Role in Maintaining Colonic Homeostasis" by BioRender.com (2020). Retrieved from: <https://app.biorender.com/biorender-templates>
- Lamm ME. Current concepts in mucosal immunity. IV. How epithelial transport of IgA antibodies relates to host defense. *Am J Physiol.* 1998;274(4):G614-G617. doi:10.1152/ajpgi.1998.274.4.g614
- Rojas R, Apodaca G. Immunoglobulin transport across polarised epithelial cells. *Nat Rev Mol Cell Biol.* 2002;3(12):944-955. doi:10.1038/nrm972. PMID: 12461560

Long Description: I g A Secretion

This diagram explains the secretion and structure of **secretory I g A (s I g A)**, an important immune component of the intestinal mucosal barrier.

- On the left, **I g A + plasma cells** in the **lamina propria** (beneath the intestinal lining) produce **I g A dimers**.
- These I g A dimers travel upward through the intestinal epithelial cells to reach the **mucosal surface**, where they are secreted.
- Once secreted, the I g A is called **secretory I g A**, and it plays a key role in neutralizing pathogens in the gut.
- On the right, a magnified image shows the structure of secretory I g A, which includes:
 - Two I g A antibodies joined by a **joint (J) chain**
 - A **secretory chain**, which stabilizes the molecule and helps it resist enzymatic degradation in the gut.
- This system enables immune protection at mucosal surfaces by preventing pathogens from attaching to and invading epithelial cells.

Fecal sIgA Testing: Significance of Increased or Decreased Levels

- **Secretory IgA is increased** in the presence of potentially harmful antigens such as **pathogenic bacteria, parasites, yeast, viruses**, abnormal cell antigens, and allergenic proteins (and should renormalize with eradication).
- **sIgA production may be decreased in cases of mental or physical stress or inadequate nutrition.**
 - Dietary restrictions, excessive alcohol intake, body mass loss, negative mood, and anxiety have been associated with lowered sIgA production.

References

FECAL SIGA

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- López MC. Chronic alcohol consumption regulates the expression of poly immunoglobulin receptor (plgR) and secretory IgA in the gut. *Toxicol Appl Pharmacol.* 2017;333:84-91. doi:10.1016/j.taap.2017.08.013
- Kang LJ, Vu KN, Koleva PT, et al. Maternal psychological distress before birth influences gut immunity in mid-infancy. *Clin Exp Allergy.* 2020;50(2):178-188. doi:10.1111/cea.13551
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Secretory IgA: High Response

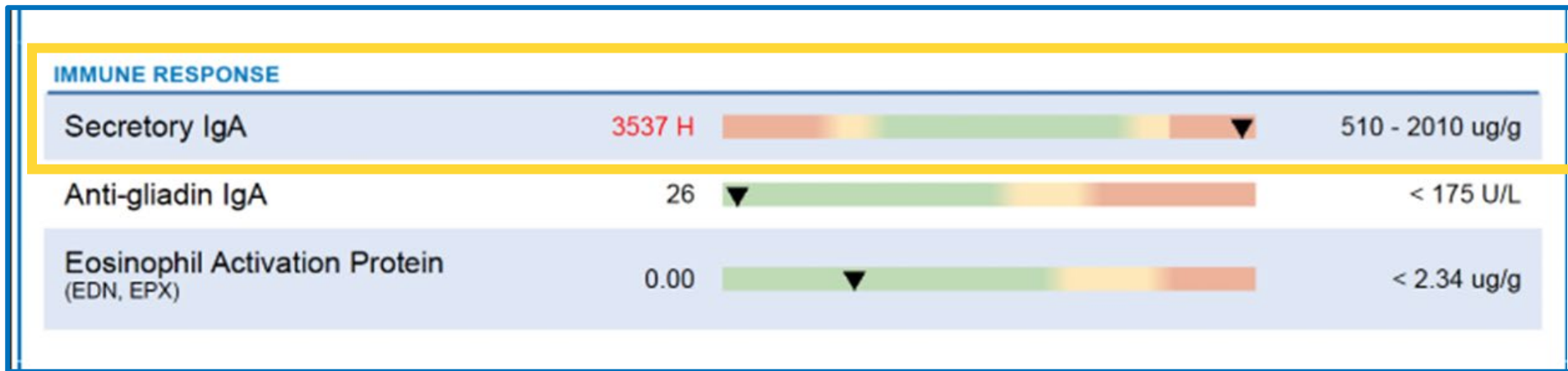


Image courtesy of Cheri Wood, ND

Secretory IgA: Low Response

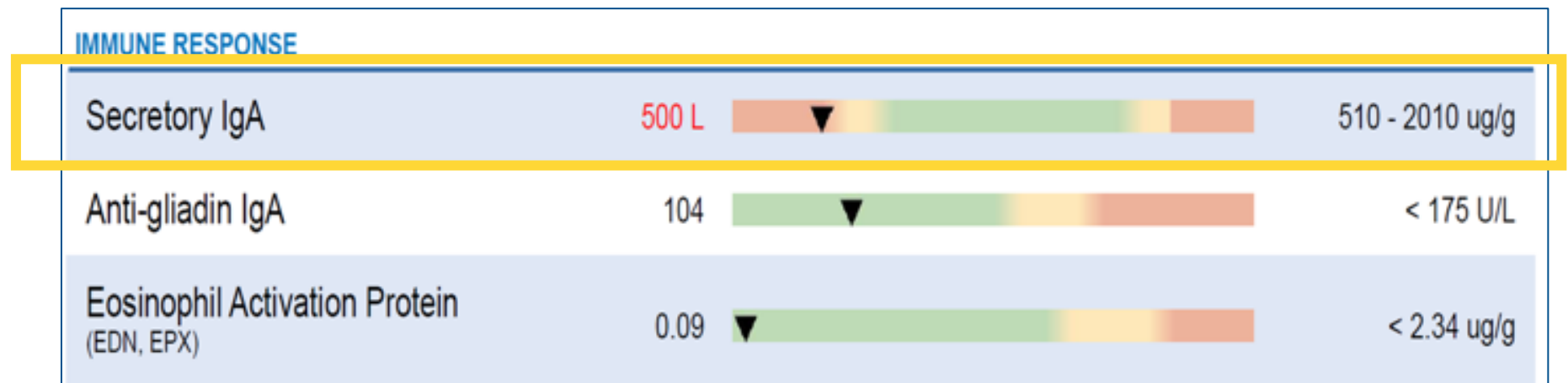


Image courtesy of Elizabeth McMillan, CNS, LDN

Lysozyme:

More research is needed

Upregulated in:

- Barrett's esophagus
- Chronic gastritis
- Gluten-induced atrophic duodenitis (celiac disease)
- Collagenous colitis
- Lymphocytic colitis
- Crohn's colitis

As a response to the special types of bacteria detected in these diseases

Mechanism:

Lysozyme upregulation protects individual mucosal segments from chronic inflammation.

Does not diagnose or determine disease activity of inflammatory bowel diseases.

High levels of fecal lysozyme (>2,000) may be suggestive of IBD

References

LYSOZYME

- Rubio CA. The natural antimicrobial enzyme lysozyme is up-regulated in gastrointestinal inflammatory conditions. *Pathogens*. 2014;3(1):73-92. doi:10.3390/pathogens3010073
- Rocha JD, Schlossmacher MG, Philpott DJ. LRRK2 and Nod2 promote lysozyme sorting in Paneth cells. *Nat Immunol*. 2015;16(9):898-900. doi:10.1038/ni.3255
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- Rubio CA. Increased production of lysozyme associated with bacterial proliferation in Barrett's esophagitis, chronic gastritis, gluten-induced atrophic duodenitis (celiac disease), lymphocytic colitis, collagenous colitis, ulcerative colitis and Crohn's colitis. *Anticancer Res*. 2015;35(12):6365-6372.
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- van der Sluys Veer A, Brouwer J, Biemond I, Bohbouth GE, Verspaget HW, Lamers CB. Fecal lysozyme in assessment of disease activity in inflammatory bowel disease. *Dig Dis Sci*. 1998;43(3):590-595. doi:10.1023/a:1018823426917

Laboratory Evaluation of Gut Dysfunction: Part 3

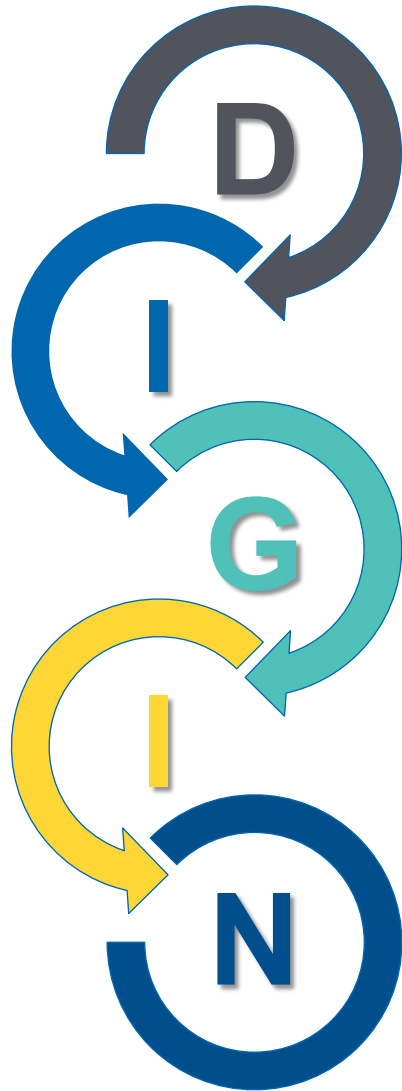
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Stool test biomarkers of the gut microbiota and dysbiosis
- Overview



Interpretation of Analytes of Gut Microbiota and Dysbiosis



Digestion/Absorption

Intestinal Permeability

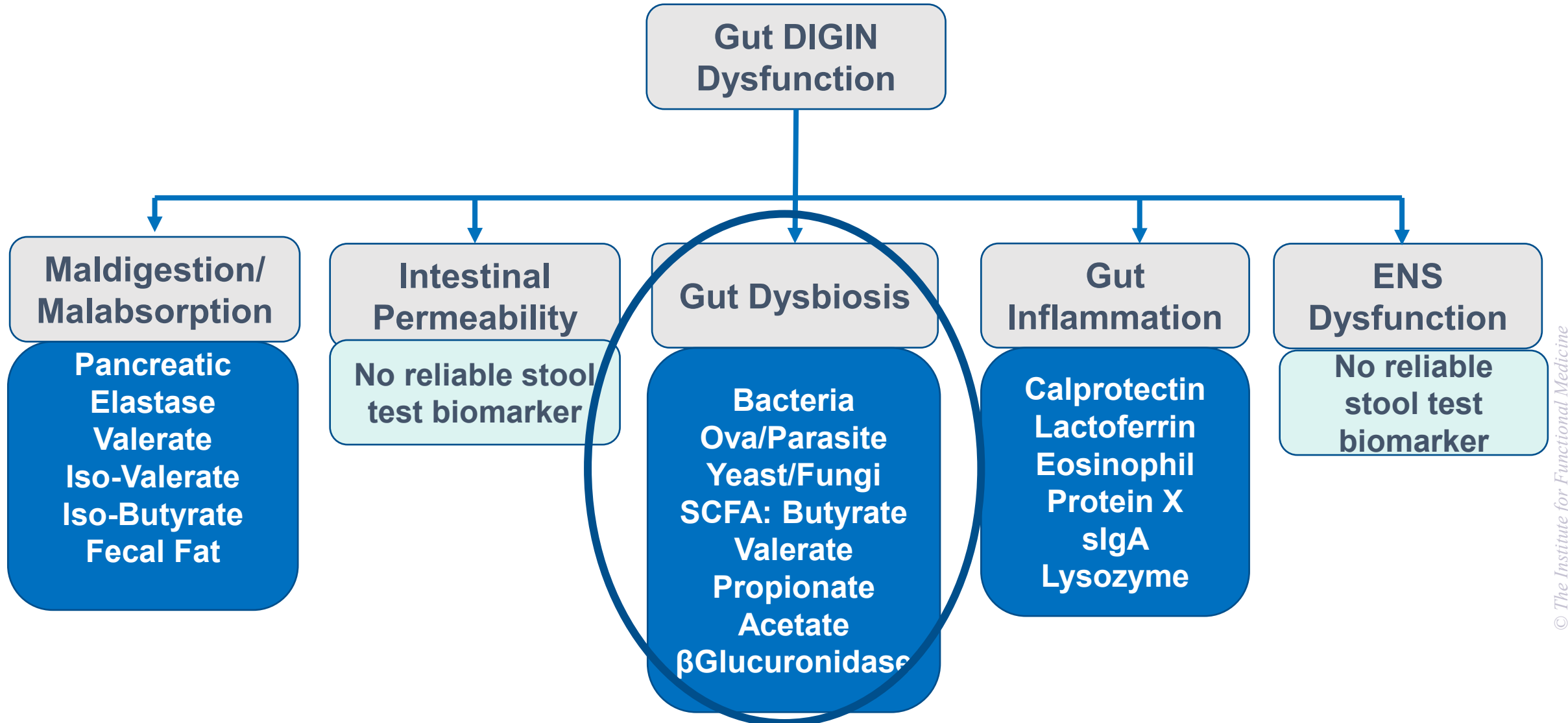
Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System (Enteric)

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

DIGIN Dysfunctions and Stool Test Biomarkers



Stool Biomarker Components

Microbiota and Its Metabolic Products:

- **Microbes**

- Category – bacteria, fungal, protozoal
- Action – beneficial, commensal, pathogen/potential pathogen

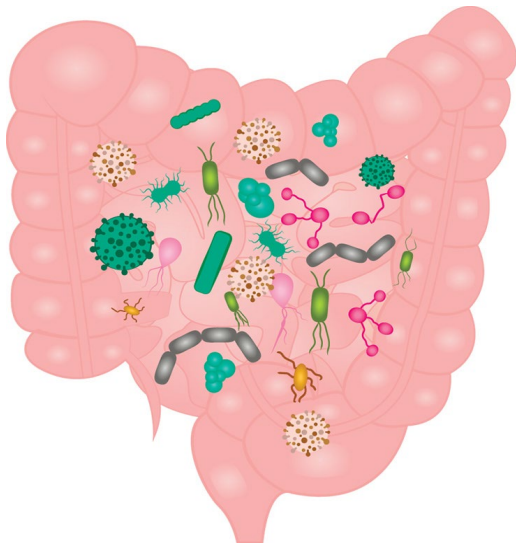
- **Metabolic Products**

- Short-chain fatty acids
- Beta-glucuronidase
- Secondary bile acids
- pH

Microbiota Testing: Clinical Context

Dysbiosis Patterns:

- Deficit
- Overgrowth
- Diversity
- Distribution



Clinical Context and Root Causes:

- Chronic GI Symptoms
- Inflammatory Bowel Disease
- Autoimmune Conditions
- Food Sensitivities/Allergies
- Neurological or Psychiatric Symptoms
- Metabolic Disorders
- Post-Antibiotic Recovery

References

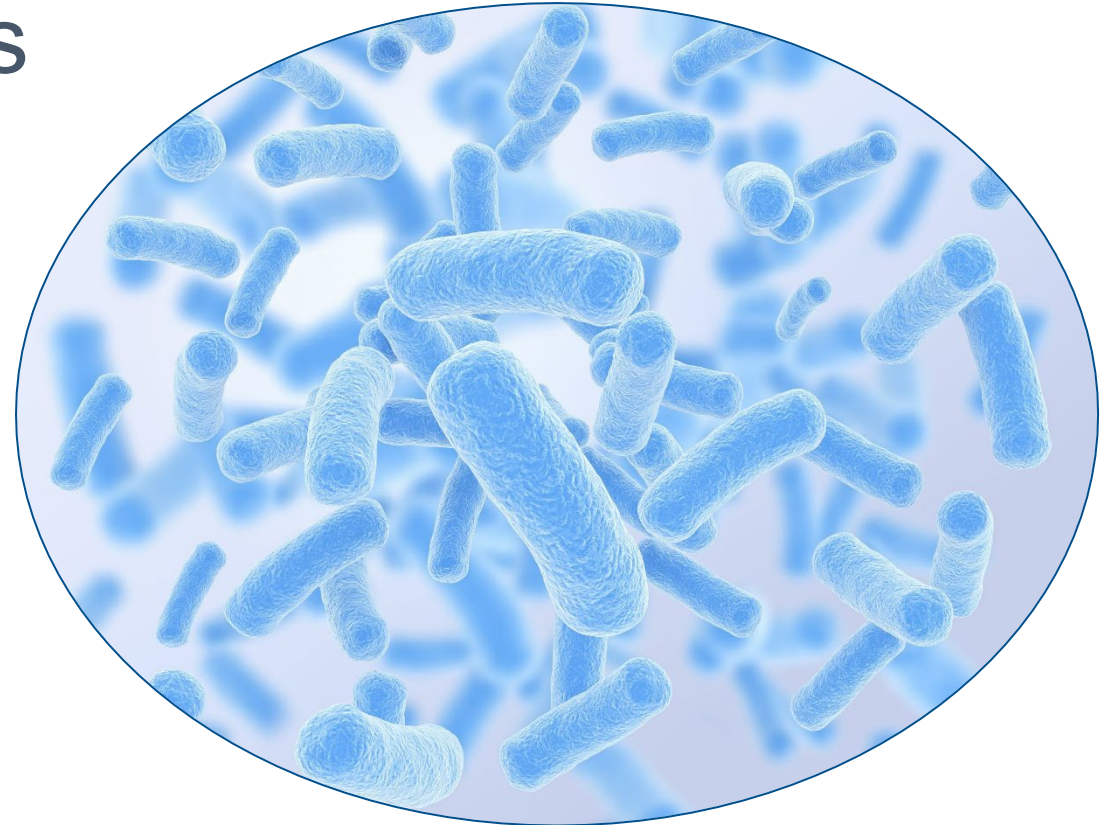
MICROBIOTA TESTING: CLINICAL CONTEXT

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- Zhao M, Chu J, Feng S, et al. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: a review. *Biomed Pharmacother.* 2023;164:114985. doi:10.1016/j.biopha.2023.114985
- Sittipo P, Lobionda S, Lee YK, Maynard CL. Intestinal microbiota and the immune system in metabolic diseases. *J Microbiol.* 2018;56(3):154-162. doi:10.1007/s12275-018-7548-y

The Microbiota: Microbial Categories

Beneficial, Commensal, Pathogens, and Potential Pathogens

- Protozoa and Helminths
- Bacteria
- Yeast



Categories of Relationship With Host

Bacteria and Yeast

- Beneficial
- Commensal
- Pathogenic
- Potentially Pathogenic/Dysbiotic



Intestinal Dysbiosis

- **A state of imbalanced microbial ecology that contributes to dysfunction and/or disease.**
- The overgrowth of micro-organisms of low intrinsic virulence induces disease by altering:
 - The nutritional status
 - The immune response
 - The elimination capacity of the host

Laboratory Evaluation of Gut Dysfunction: Part 4

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Stool test biomarkers of the gut microbiota and dysbiosis
- Parasites



Parasites: Laboratory Detection Methods

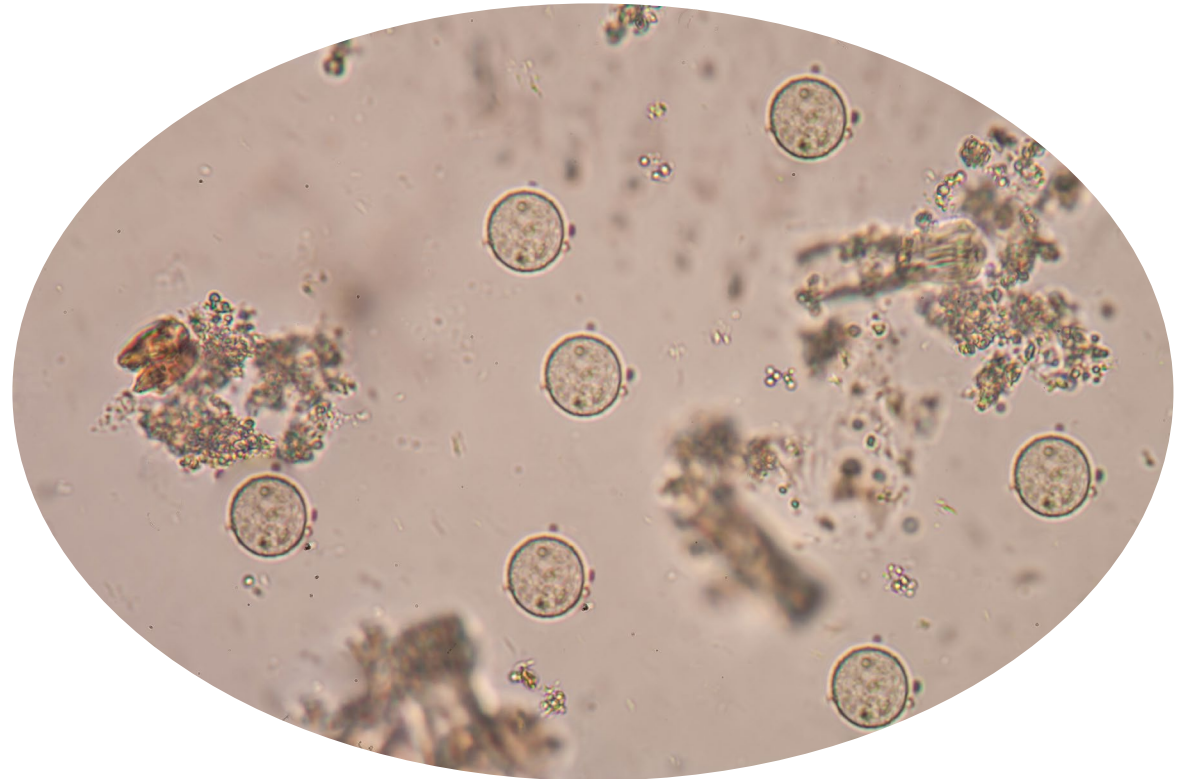
O&P Microscopy

- Traditional method, well established
- Individual samples
- Pooled samples

EIAs (most common)

- *G. lamblia*
- *Cryptosporidium*
- *E. histolytica*

PCR Probes (emerging)



Parasites: Gold Standard = Microscopy

Microscopy detection rates are a function of:

- Specimen collection and handling
- Number and kind of specimens examined
- Concentration procedures
- Staining procedures
- Macroscopic and microscopic examination techniques
- Quality of training, frequency of practice, and dedication of laboratory personnel

Parasites: Laboratory Report Example

Genus/species	Result
Nematodes - roundworms	
<i>Ancylostoma/Necator</i> (Hookworm)	Not Detected
<i>Ascaris lumbricoides</i>	Not Detected
<i>Capillaria philippinensis</i>	Not Detected
<i>Enterobius vermicularis</i>	Not Detected
<i>Strongyloides stercoralis</i>	Not Detected
<i>Trichuris trichiura</i>	Not Detected
Cestodes - tapeworms	
<i>Diphyllobothrium latum</i>	Not Detected
<i>Dipylidium caninum</i>	Not Detected
<i>Hymenolepis diminuta</i>	Not Detected
<i>Hymenolepis nana</i>	Not Detected
<i>Taenia</i> spp.	
Trematodes - flukes	
<i>Clonorchis/Opisthorchis</i> spp.	
<i>Fasciola</i> spp./ <i>Fasciolopsis buski</i>	
<i>Heterophyes/Metagonimus</i>	
<i>Paragonimus</i> spp.	
<i>Schistosoma</i> spp.	
Protozoa	
<i>Balantidium coli</i>	
<i>Blastocystis</i> spp.	
<i>Chilomastix mesnili</i>	
<i>Cryptosporidium</i> spp.	
<i>Cyclospora cayetanensis</i>	
<i>Dientamoeba fragilis</i>	
<i>Entamoeba coli</i>	
<i>Entamoeba histolytica/dispar</i>	
<i>Entamoeba hartmanni</i>	
<i>Entamoeba polecki</i>	Not Detected
<i>Endolimax nana</i>	Not Detected
<i>Giardia</i>	Not Detected
<i>Iodamoeba buetschlii</i>	Not Detected
<i>Cystoisospora</i> spp.	Not Detected
<i>Trichomonads</i> (e.g. <i>Pentatrichomonas</i>)	Not Detected
Additional Findings	
White Blood Cells	Not Detected
Charcot-Leyden Crystals	Not Detected
Other Infectious Findings	

Parasitology					
PCR Parasitology - Protozoa				Methodologies: DNA by PCR	
Organism	Result	Units			Expected Result
<i>Blastocystis</i> spp.	<2.14e2	femtograms/microliter C&S stool	Not Detected		Not Detected
<i>Cryptosporidium parvum/hominis</i>	<1.76e2	genome copies/microliter C&S stool	Not Detected		Not Detected
<i>Cyclospora cayetanensis</i>	<2.65e2	genome copies/microliter C&S stool	Not Detected		Not Detected
<i>Dientamoeba fragilis</i>	<1.84e2	genome copies/microliter C&S stool	Not Detected		Not Detected
<i>Entamoeba histolytica</i>	<9.64e1	genome copies/microliter C&S stool	Not Detected		Not Detected
<i>Giardia</i>	<1.36e1	genome copies/microliter C&S stool	Not Detected		Not Detected

Images courtesy of Dan Lukaczer, ND

Parasites:

Laboratory Report Example

PARASITES		
PROTOZOA	Result	Reference
<i>Blastocystis hominis</i>	<dl	< 2.00e3
<i>Chilomastix mesnili</i>	<dl	< 1.00e5
<i>Cyclospora</i> spp.	<dl	< 5.00e4
<i>Dientamoeba fragilis</i>	<dl	< 1.00e5
<i>Endolimax nana</i>	<dl	< 1.00e4
<i>Entamoeba coli</i>	<dl	< 5.00e6
<i>Pentatrichomonas hominis</i>	<dl	< 1.00e2
WORMS		
<i>Ancylostoma duodenale</i>	Not Detected	Not Detected
<i>Ascaris lumbricoides</i>	Not Detected	Not Detected
<i>Necator americanus</i>	Not Detected	Not Detected
<i>Trichuris trichiura</i>	Not Detected	Not Detected
<i>Taenia</i> spp.	Not Detected	Not Detected

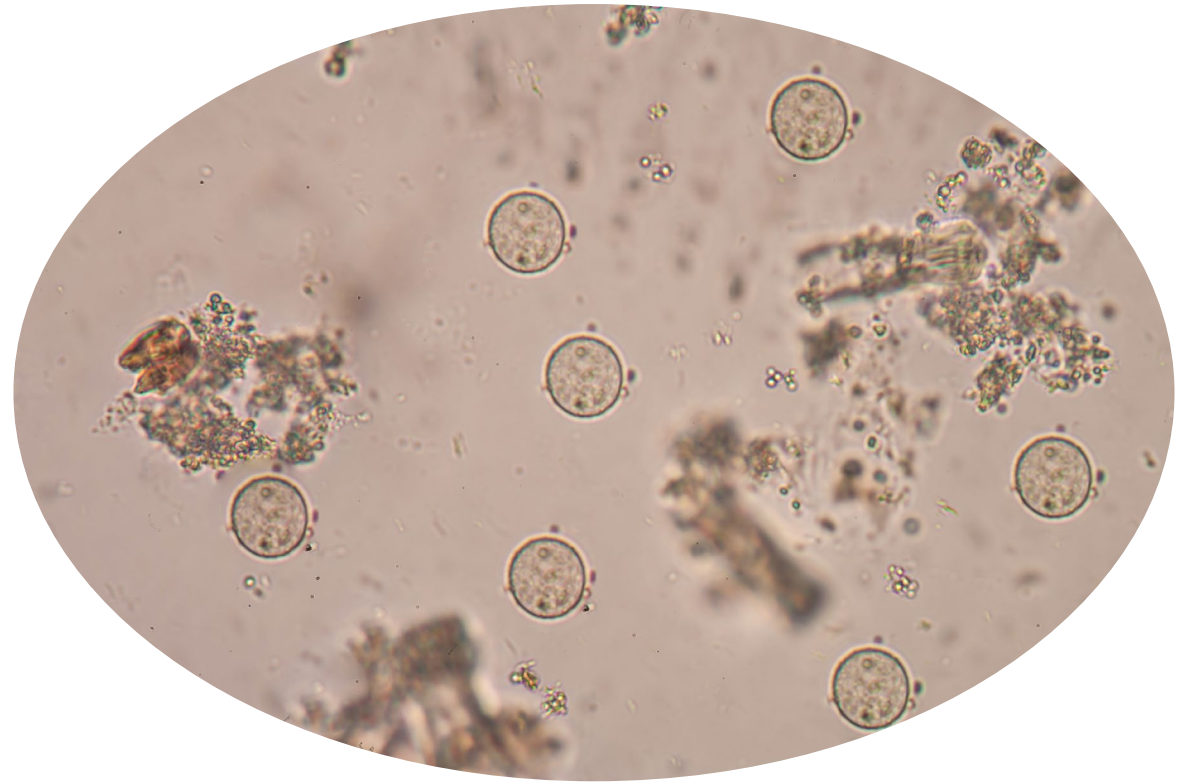
Clinical Focus

- Stool test biomarkers of the gut microbiota and dysbiosis
- Bacteria



Bacteria: Laboratory Detection Methods

- Culture
- 16S rRNA
- Metagenomics
- Metabolomics
- Diversity Indices

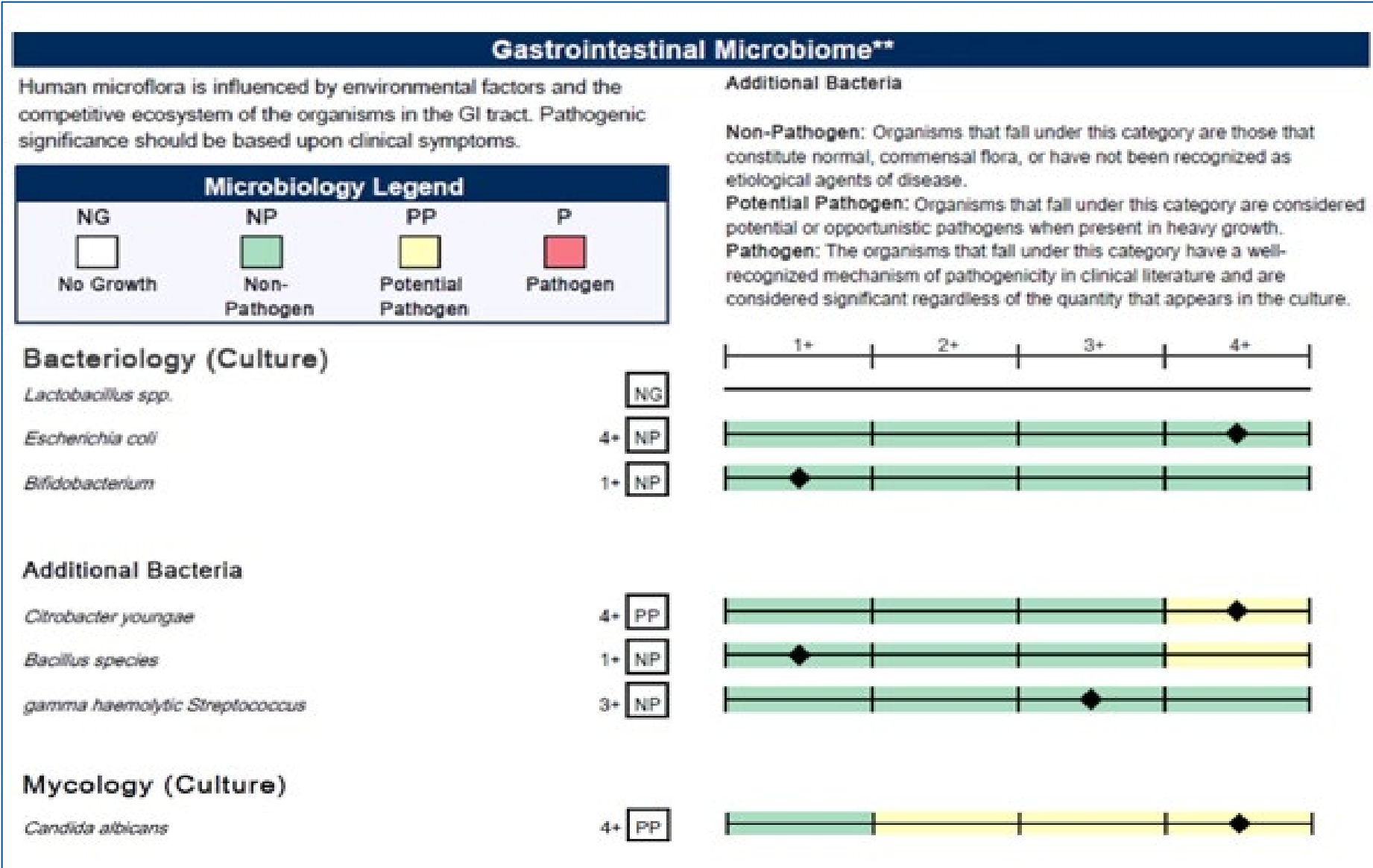


Culture

- Limited number of bacteria that can be grown
- Most organisms are anaerobic

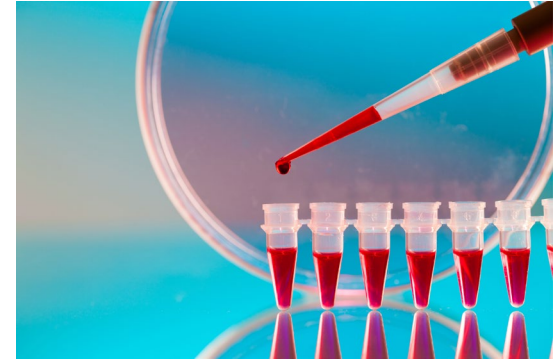


Laboratory Evaluation: Bacteria [and Yeast] Culture

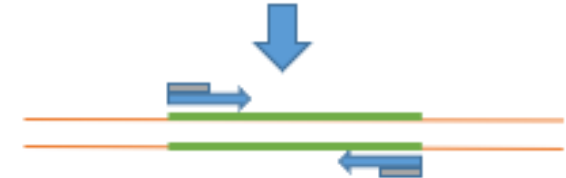


16S rRNA Gene Sequencing

- A common method for identifying bacteria is analyzing the sequence of the gene coding for 16S ribosomal RNA
- Can only identify bacteria to genus level



Purify DNA from biofilm



PCR amplify 16S rRNA gene

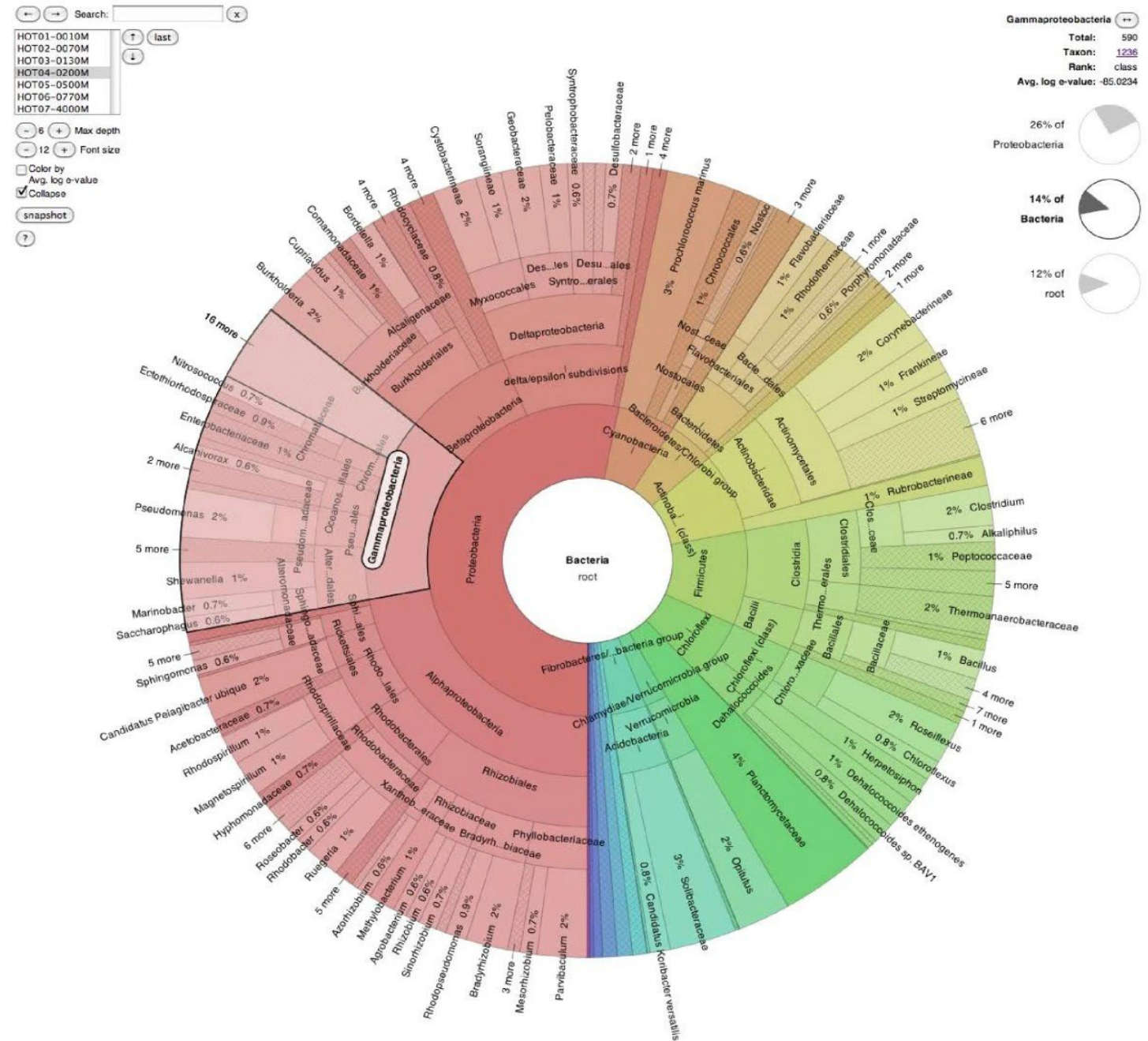


Sequence PCR product

Compare sequence to database

Identify bacterial species

Metagenomics



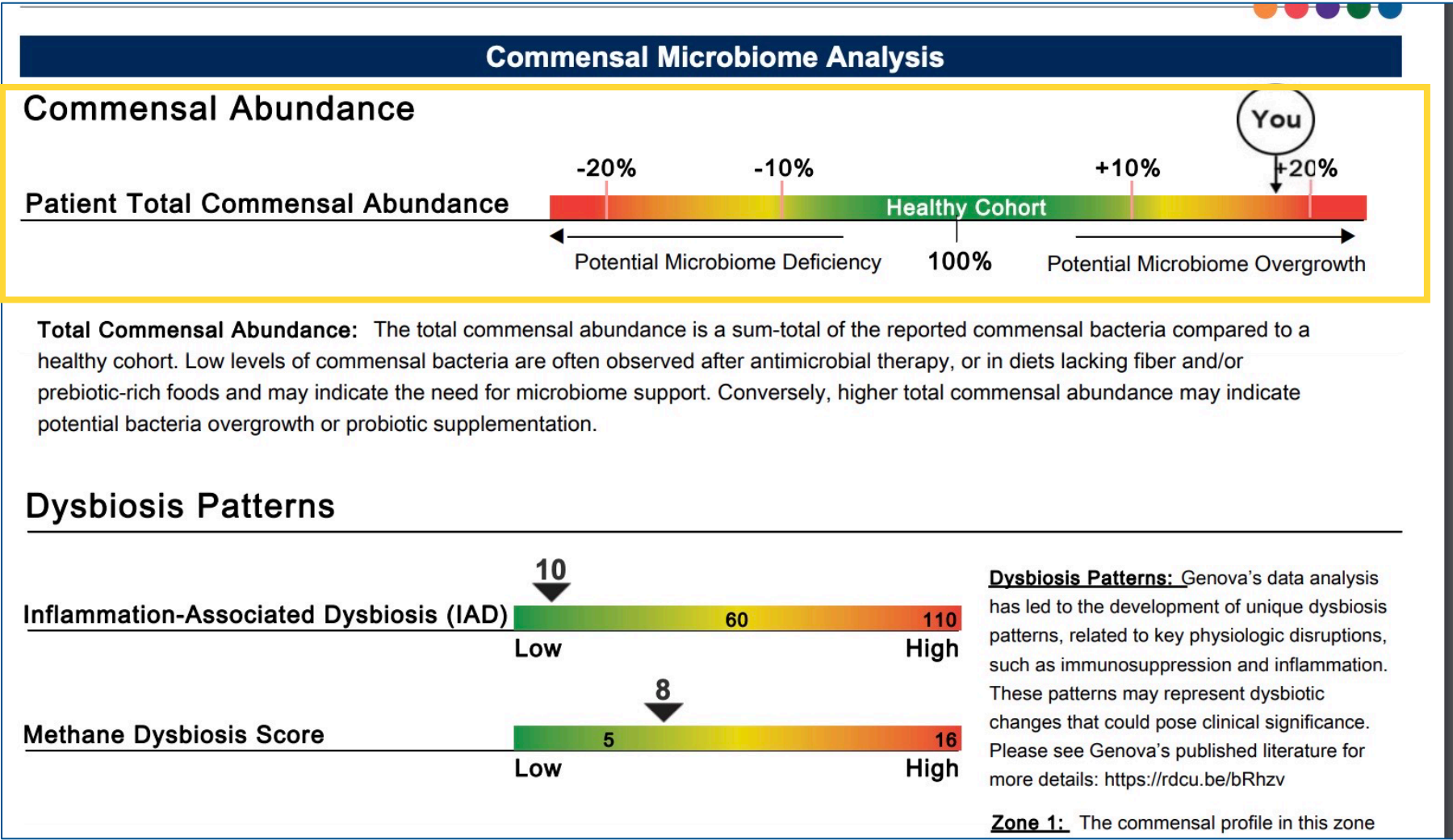
Metagenomics vs Metabolomics

Metagenomics = Composition of the gut flora (i.e., census taking)

Metabolomics = The metabolic activity of the microbiota

**The composition of the microbiota determines the metabolic activity and the metabolites.
Many beneficial functions of the microbiota involve multiple species and explain the benefits of diversity.**

Diversity Indices



Quantify the Diversity

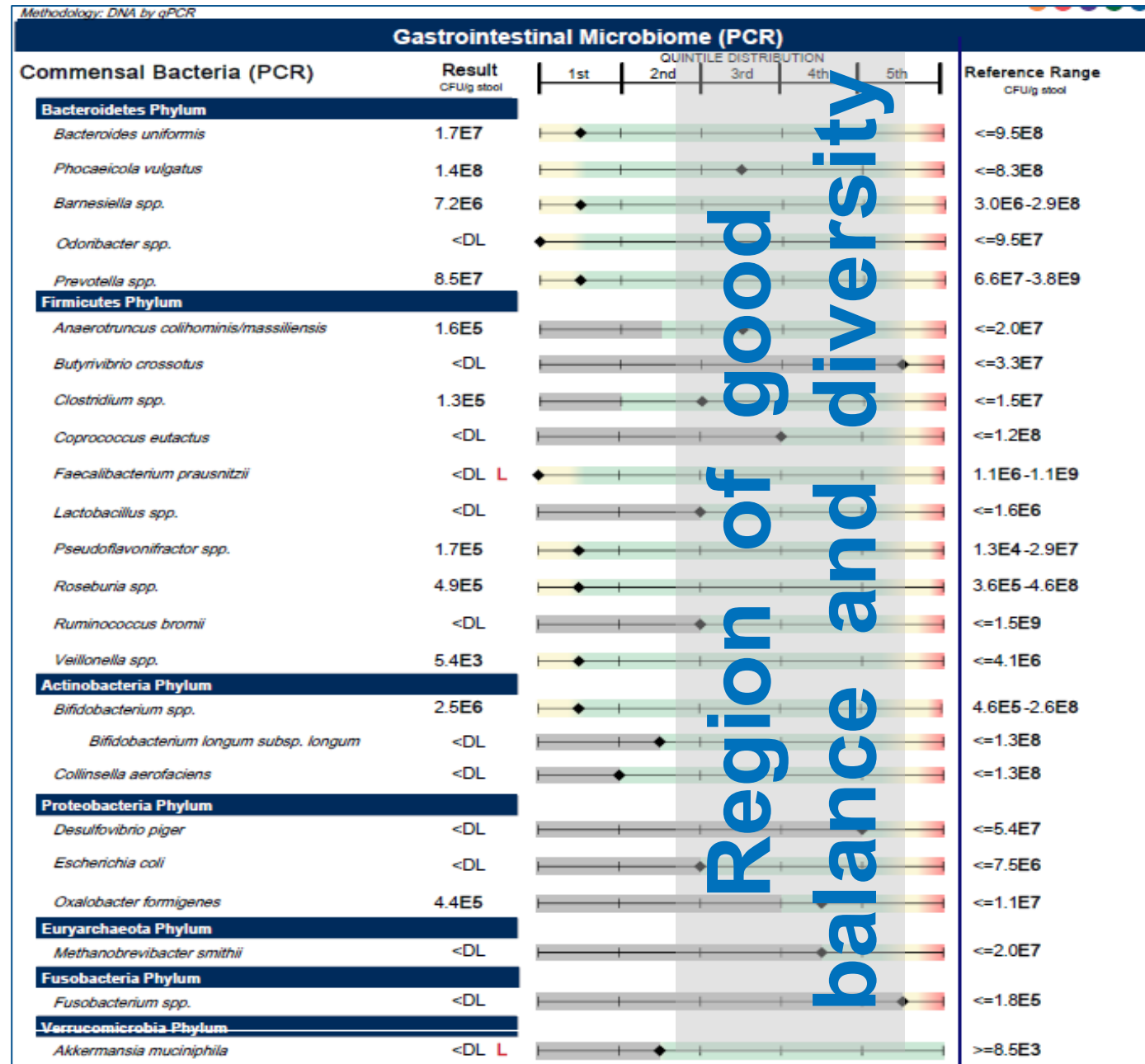


Image courtesy of Dan Lukaczer, ND

Lab Summary

Test Type	What It Measures	Strengths	Limitations
Culture-Based	Viable microbes grown in lab	Shows what's alive & susceptibility	Misses many species, esp. anaerobes
16S rRNA PCR	Bacterial DNA (16S gene)	Faster, cost-effective snapshot of bacteria	Only bacteria, limited taxonomic resolution
Metagenomics	All microbial DNA (bacteria, viruses, fungi)	Comprehensive identity & genetic capability	Expensive, complex data
Metabolomics	Metabolites (SCFA, bile acids, etc.)	Reveals microbial function	Doesn't show who made what
Diversity Index	Microbial variance from microbiome profile	Concise overview of microbial abundance and balance	Calculated score from lab

- Yin X, Altman T, Rutherford E, et al. A comparative evaluation of tools to predict metabolite profiles from microbiome sequencing data. *Front Microbiol.* 2020;11:595910. doi:10.3389/fmicb.2020.595910
- Muller E, Algavi YM, Borenstein E. A meta-analysis study of the robustness and universality of gut microbiome-metabolome associations. *Microbiome.* 2021;9(1):203. doi:10.1186/s40168-021-01149-z
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Laboratory Evaluation of Gut Dysfunction: Part 5

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Stool test biomarkers of the gut microbiota and dysbiosis
- Microbial metabolites



Stool Biomarker Components

Microbiota and its metabolic products:

- **Microbes**

- Category – bacteria, fungal, protozoal
- Action – beneficial, commensal, pathogen/potential pathogen

- **Metabolic Products:**

- Short-chain fatty acids
- Beta-glucuronidase
- Secondary bile acids
- pH

Short-Chain Fatty Acids (SCFAs)

- Major SCFAs include **acetate, propionate, and butyrate**.
 - Produced during the fermentation of non-digestible polysaccharides by gut microbiota.
 - Serve as an important source of energy for colonocytes, liver cells, and skeletal muscle.
 - Play a role in gut motility, intestinal barrier permeability, and immune function.

Short-Chain Fatty Acids (SCFAs)

- Major SCFAs include **acetate, propionate, and butyrate.**
 - Increase sensation of satiety and have been shown to reduce food consumption.
 - Reduce human colon cancer cell growth.
 - Inhibit production of inflammatory cytokines in multiple tissue types.

Short-Chain Fatty Acids (SCFAs)

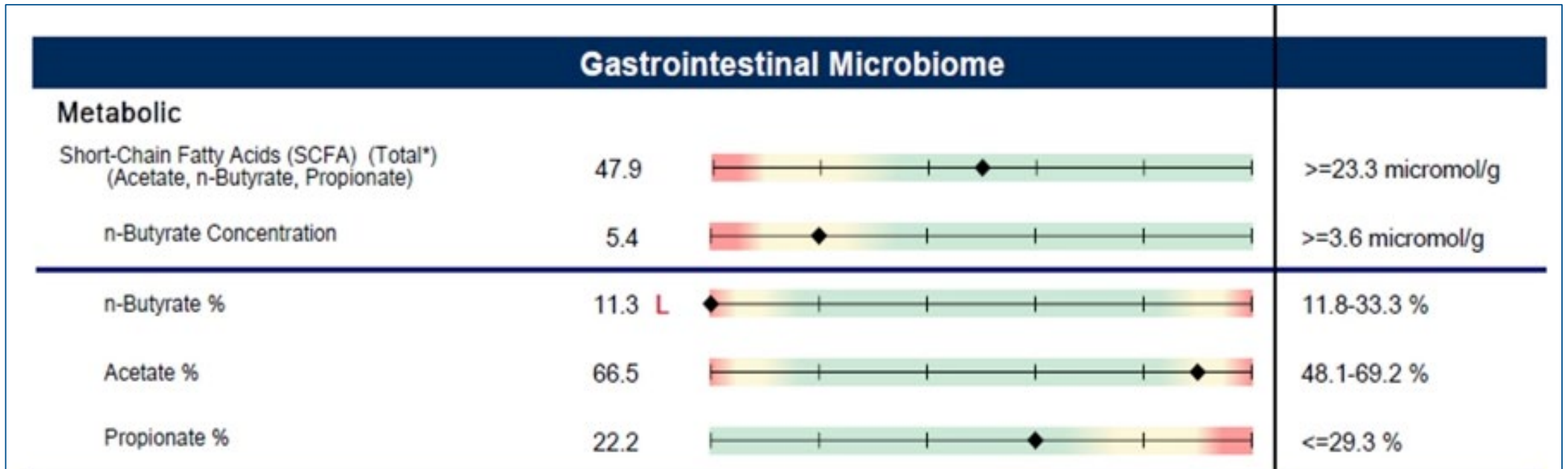
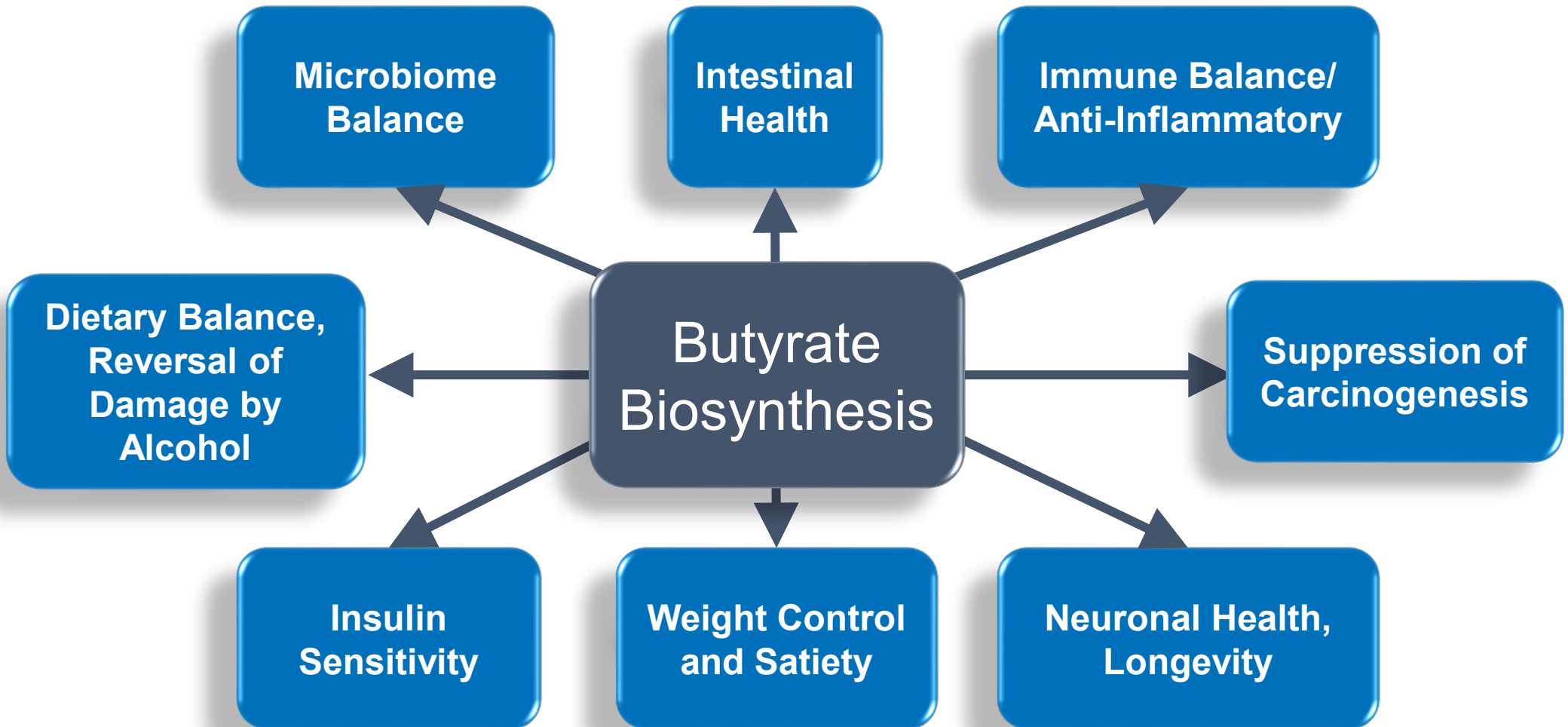


Image courtesy of Elizabeth Boham, MD, MS, RD

Butyrate

Beneficial Functions



Short-Chain Fatty Acids

Gut Microbiome Metabolites

Metabolic

Short-Chain Fatty Acids (SCFA) (Total*)
(Acetate, n-Butyrate, Propionate)

16.5 L



≥ 23.3 micromol/g

n-Butyrate Concentration

2.2 L



≥ 3.6 micromol/g

n-Butyrate %

13.3



11.8-33.3 %

Acetate %

60.5



48.1-69.2 %

Propionate %

26.1



≤ 29.3 %

Beta-glucuronidase

758



368-6,266 U/g

β-Glucuronidase

- Lysosomal enzyme involved in the degradation of glucuronate-containing glycosaminoglycan.
- Normally produced in the intestine and liver in relatively low levels. May mediate dysfunction when elevated.
- The major microbial producers of β-glucuronidase are colonic bacteria, including *E. coli*, *Clostridium paraputrificum*, *Clostridium clostridioforme*, *Clostridium perfringens*, *Bacteroides fragilis*, *Bacteroides vulgatus*, *Bacteroides uniformis*, *Ruminococcus gnavus*, *Peptostreptococcus*, *Staphylococcus*, and *Eubacterium*.

β -Glucuronidase

- Increased activity is stimulated by tobacco, exposure to toxic substances and carcinogens, consumption of red meat, and antibiotics.
- Putative Mechanism: Intestinal uncoupling of glucuronide moiety bound to toxins (during phase II liver biotransformation). Toxins may then be reabsorbed rather than eliminated.

β -Glucuronidase

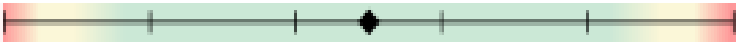
Gut Microbiome Metabolites				
Metabolic				
Short-Chain Fatty Acids (SCFA) (Total*) (Acetate, n-Butyrate, Propionate)	45.1			≥ 23.3 micromol/g
n-Butyrate Concentration	9.2			≥ 3.6 micromol/g
n-Butyrate %	20.4			11.8-33.3 %
Acetate %	58.5			48.1-69.2 %
Propionate %	21.1			≤ 29.3 %
Beta-glucuronidase	14,454 H			368-6,266 U/g

Image courtesy of Elizabeth Boham, MD, MS, RD

Secondary Bile Acids

- Secondary bile acids (SBAs) are formed by bacterial metabolism of primary bile acids in the colon.
- SBAs are increased by dietary factors, primarily red meat and saturated fats.
- Due to their hydrophobic nature, elevated levels of SBAs may cause damage to cell membranes, resulting in destruction of intestinal epithelium.

Secondary Bile Acids

- Elevated levels of secondary bile acids are associated with an increase in inflammatory cytokines in colonic mucosa.
- Increased levels of secondary bile acids, e.g., deoxycholic acid and lithocholic acid, are associated with increased incidence of colorectal cancer.

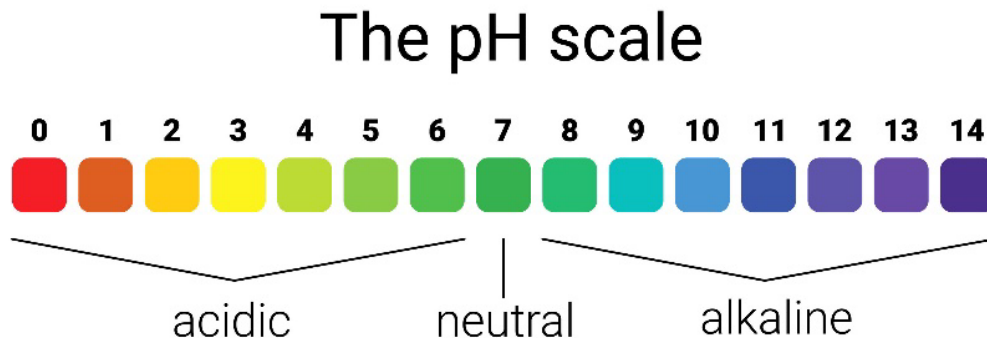
Fecal pH: Non-Specific Biomarker

Low pH

- CHO malabsorption
- CHO maldigestion
- Fast transit
- Organic acids
- SIBO

High pH

- High-protein and/or low-fiber diet
- Dysbiosis
- Slow transit time
- Hypochlorhydria
- Pancreatic bicarbonate
- Associated with increased risk for colorectal cancer



References

FECAL PH

Low pH References:

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Metabolic Products Testing: Clinical Context

Biomarker	Clinical Context if High	Clinical Context if Low	Root Cause of Imbalance
Short-Chain Fatty Acids	Overfermentation, SIBO	Gut inflammation, low fiber, dysbiosis	• Low-fiber diet • Dysbiosis/SIBO • Antibiotic use • Rapid transit time
Beta Glucuronidase	Toxin reabsorption, estrogen dominance	Lower microbial activity (rarely a concern)	• Dysbiosis • Diet • Liver burden
Secondary Bile Acids	Colon cancer risk, inflammation	Decreased fat digestion and metabolism	• Dysbiosis • Liver/gallbladder issues • Diet
Fecal pH	Alkaline: protein putrefaction, bile issues	Acidic: fermentation, SIBO	• Diet • SIBO • Malabsorption • Microbial imbalance

References

METABOLIC PRODUCTS TESTING: CLINICAL CONTEXT

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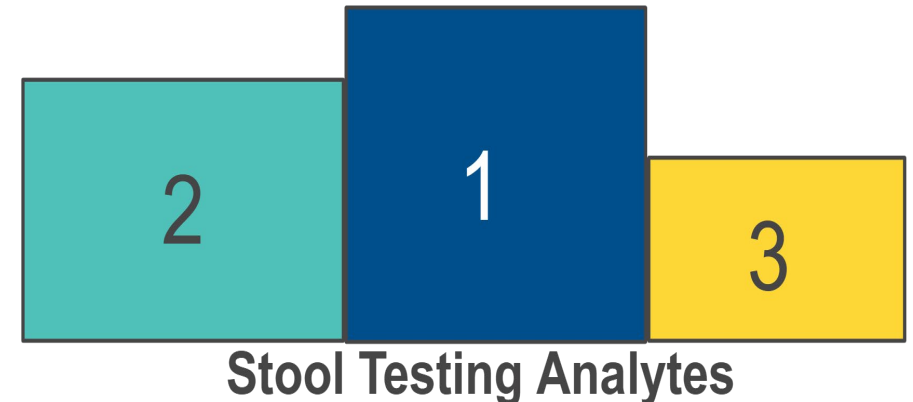
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Stool Biomarker Components

Additional stool analytes that have lower validity and utility

- Fecal zonulin vs serum zonulin
- Anti-gliadin IgA

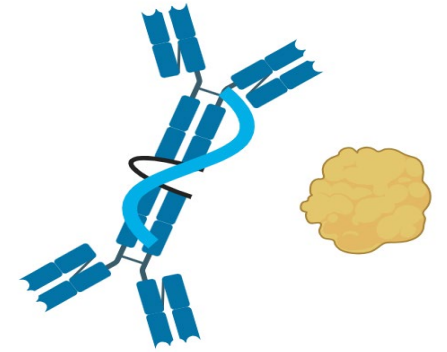


Zonulin Testing:

Increased Intestinal Permeability Testing

- **The permeability of tight junctions is regulated by zonulin and occludin proteins**, intracellular proteins that connect the tight junctions to the cytoskeleton of the adjacent cells.
- **Due to its fluctuation, a single measurement of fecal zonulin level is not recommended for assessment of intestinal barrier integrity.**
 - Its sustained serological half-life is not well understood.
- Measurement of serum IgG and IgA antibodies against zonulin, occludin, and other tight junction proteins is proposed for the evaluation of the loss of intestinal barrier integrity.

Anti-Gliadin IgA in Stool



- Anti-gliadin antibodies (AGAs) are immune proteins produced in response to gliadin, a component of gluten.
- Stool testing for AGAs is not standard clinical practice, and the American College of Gastroenterology and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition do not endorse stool AGA testing for diagnosis.

If gluten-related disorders are suspected, blood-based tests and clinical evaluation remain the gold standard.

Clinical Focus

- Conclusion and Key Takeaways
- A Look Ahead



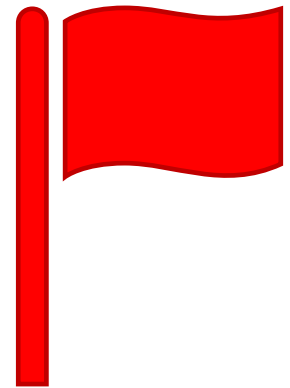
Key Takeaways



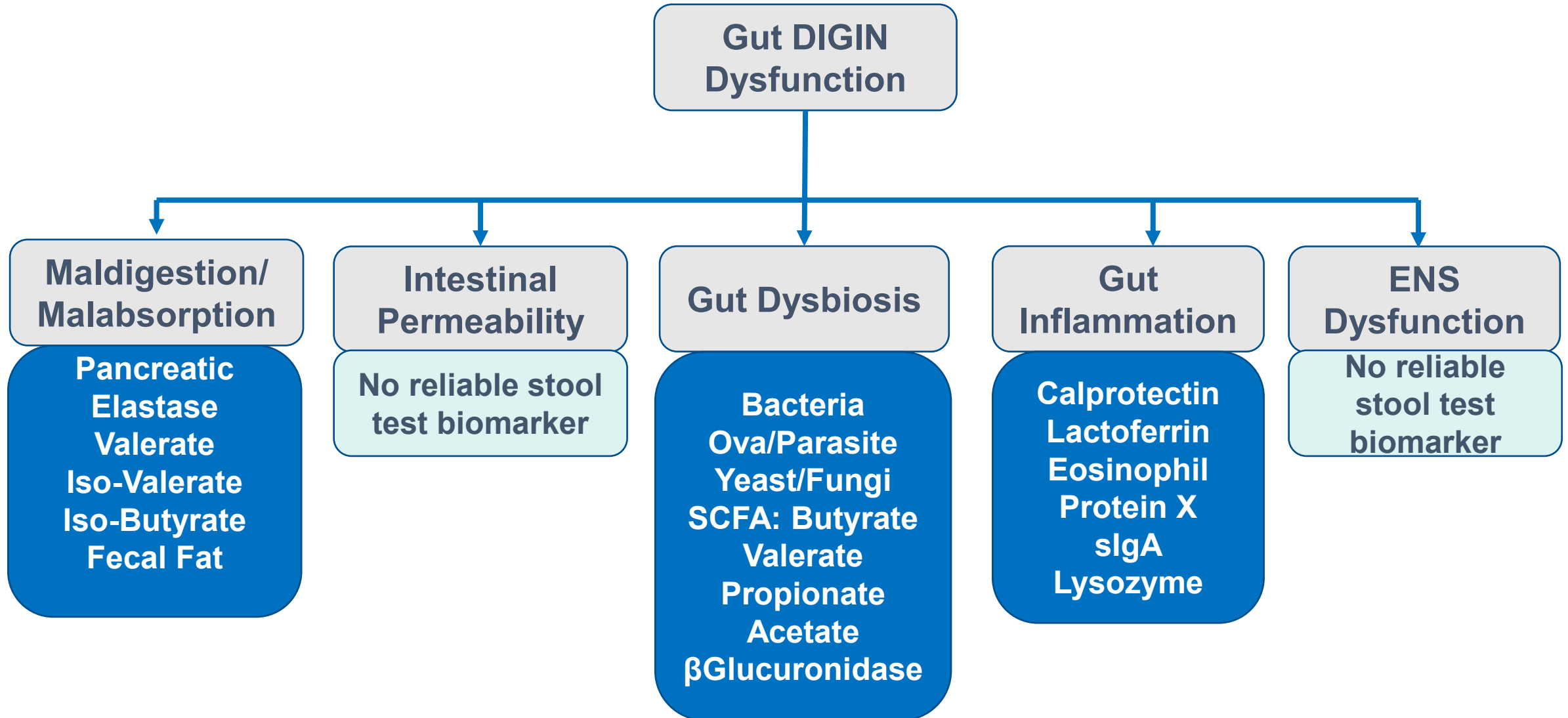
Key Takeaways & Red Flags

IDENTIFY RED FLAGS that may require urgent referral:

- Weight loss
- Fever
- Melena, hematochezia, blood per rectum
- Tachycardia
- Hypotension



DIGIN Dysfunctions and Stool Test Biomarkers



Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

This graphic categorizes stool test biomarkers associated with key D I G I N gut dysfunctions and is offered as a review of the same slide presented earlier in this talk.

- **Maldigestion/Malabsorption**

Associated stool biomarkers:

- Pancreatic elastase
- Valerate
- Iso-valerate
- Iso-butyrate
- Fecal fat

- **Intestinal Permeability**

There is **no reliable stool test biomarker** currently available for this dysfunction.

- **Gut Dysbiosis**

Associated stool biomarkers:

- Bacteria
- Ova/parasites
- Yeast/fungi
- Short-chain fatty acids (S C F A): butyrate, valerate, propionate, acetate
- Beta-glucuronidase

- **Gut Inflammation**

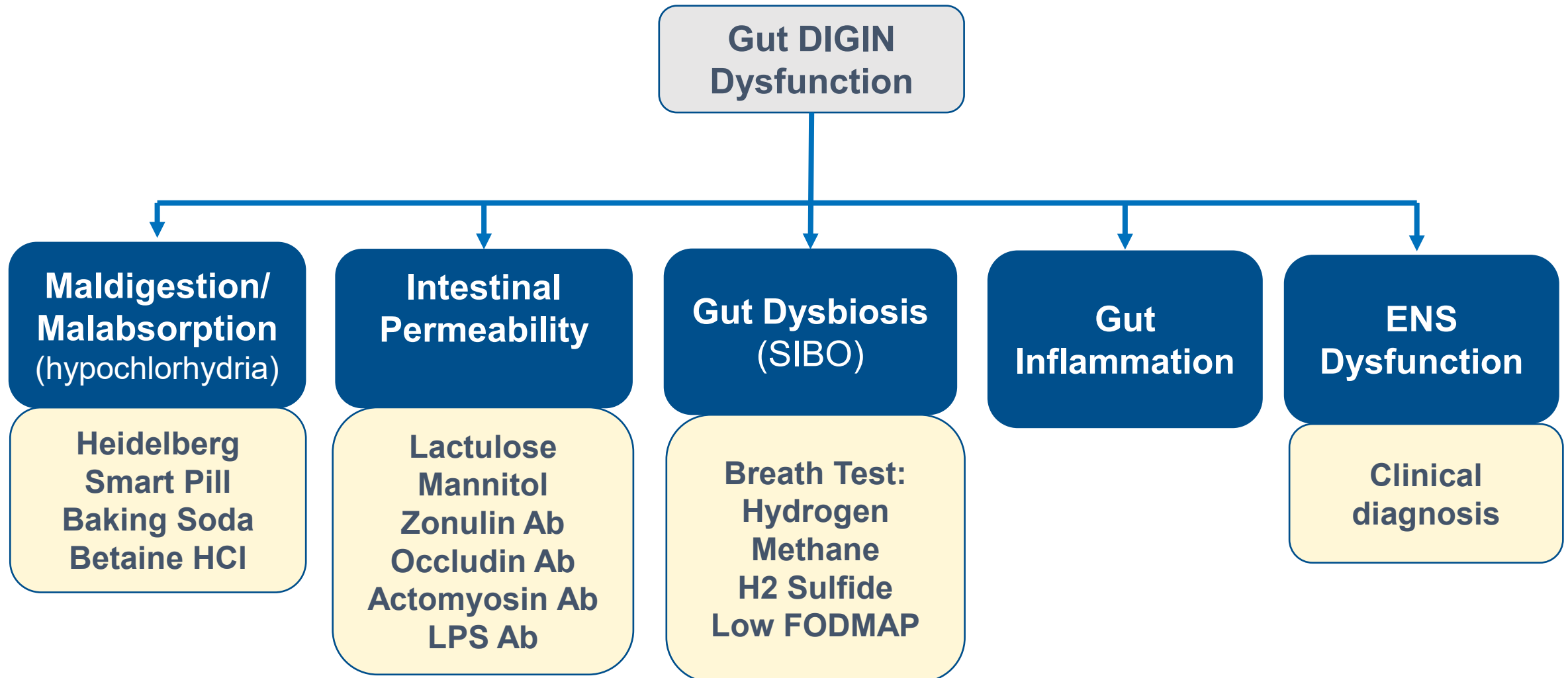
Associated stool biomarkers:

- Calprotectin
- Lactoferrin
- Eosinophil protein X
- Secretory I g A (s I g A)
- Lysozyme

- **Enteric Nervous System (E N S) Dysfunction**

There is **no reliable stool test biomarker** currently available for this dysfunction.

DIGIN Dysfunctions Evaluated by Diagnostic Modalities Other Than Stool Test



Long Description: D I G I N Dysfunctions

- This graphic outlines diagnostic tools used to assess gut DIGIN dysfunctions beyond standard stool testing.
- **Maldigestion/Malabsorption (hypochlorhydria)**
Assessed with:
 - Heidelberg test
 - Smart Pill
 - Baking soda test
 - Betaine HCl challenge
- **Intestinal Permeability**
Assessed with:
 - Lactulose and mannitol permeability tests
 - Zonulin antibody (Ab)
 - Occludin antibody
 - Actomyosin antibody
 - Lipopolysaccharide (LPS) antibody
- **Gut Dysbiosis (Small Intestinal Bacterial Overgrowth – S I B O)**
Assessed with:
 - Breath tests measuring hydrogen, methane, and hydrogen sulfide
 - Low FODMAP dietary response
- **Gut Inflammation**
No specific tests are listed in this graphic, implying other tools may be used or it is assessed clinically or with lab markers not included here.
- **ENS (Enteric Nervous System) Dysfunction**
Assessed by **clinical diagnosis**, without specific biomarker testing

Stool Test Biomarkers/Lab Analytes

GI	Associated Biomarkers
Digestion/Absorption	• Pancreatic elastase • Valerate • Isovalerate • Iso-butyrate • Fecal fat
Inflammation	• Calprotectin • Lactoferrin • Eosinophil protein X • Secretory IgA (sIgA) • Lysozyme

Stool Test Biomarkers/Lab Analytes

GI	Associated Biomarkers
Microbiome: Organisms	• Ova and parasite (protozoa & helminths) • Bacteria • Viruses • Fungal, yeast, candida
Microbiome: Metabolites	• Short-chain fatty acids (SCFAs) • Butyrate • Acetate • Propionate • Valerate • Beta glucuronidase • Fecal pH

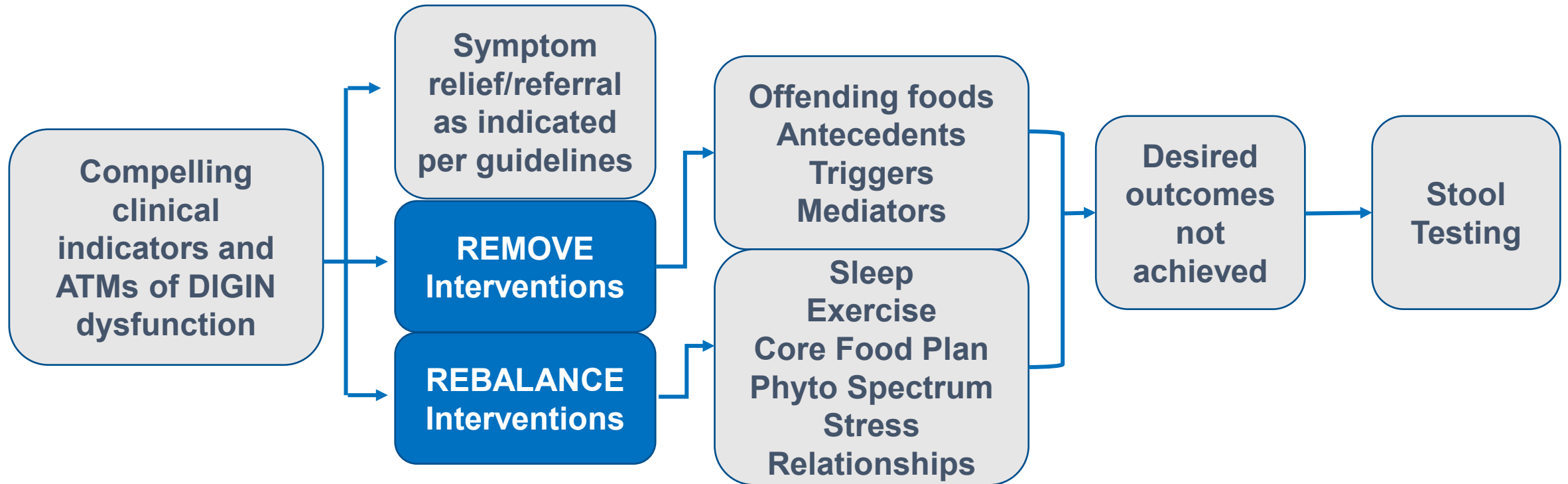
Key Takeaways



DIGIN Dysfunctions

	GUT DYSFUNCTION	RELIABLE SYMPTOMS	ASSOCIATED CONDITIONS	CORRELATED STOOL TEST BIOMARKERS	OTHER TESTING OPTIONS
D	Digestion Absorption	Immediate PP Symptoms	Nutrient Deficiencies PPI; Low HCl	Pancreatic elastase; Fecal fat; Stool pH; Iso/valerate; Isobutyrate	Baking soda, Betaine HCl, Nutrient levels
I	Intestinal Permeability	Few +/- Food Reactions	Inflammatory Conditions; Autoimmunity	None/low validity	Lactulose/mannitol Serum antibodies
G	Dysbiosis SIBO/SIMO	+/- Immediate PP Symptoms	Abx; PPI Low HCl	Same as above if maldigestion occurs	Breath tests
G	Dysbiosis Colon	Few +/- Delayed PP Symptoms	Many Chronic Diseases	Bacteria; Yeast; Parasites; SCFA; Butyrate; B-gluc.; Sec. BAs; pH; Urine OAs	Organic acids
I	Gut Inflammation	Few +/- Bleeding	Inflammatory Conditions; IBD	Calprotectin; Lactoferrin; Eosinophil protein X; Secretory IgA (sIgA)	Endoscopy
N	ENS	Few +/- Dysmotility	Stress	None/low validity	Clinical diagnosis

Gut Dysfunction Empirical Treatment Decision Tree



Long Description: Gut Dysfunction Empirical Treatment Decision Tree

This diagram presents a clinical decision tree for empirically treating gut dysfunction based on DIGIN-related signs and symptoms.

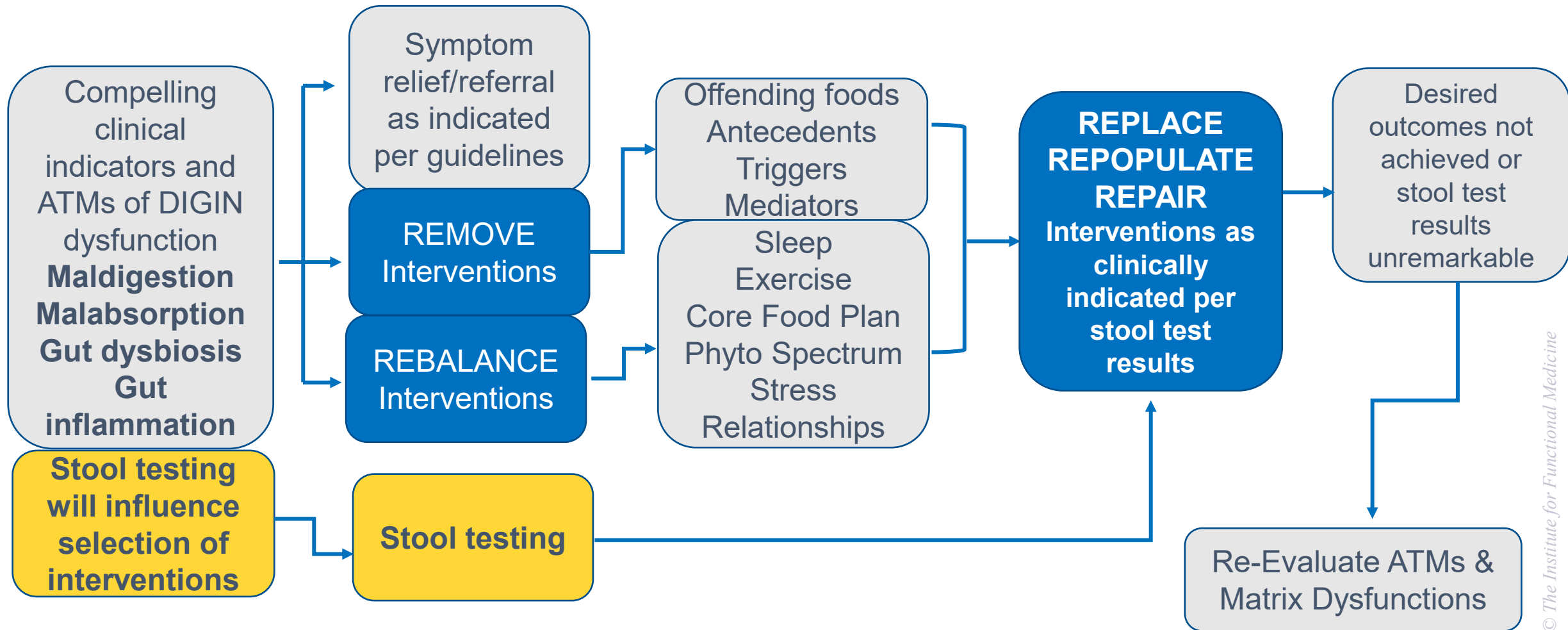
- The process begins with **compelling clinical indicators** and **ATMs (Antecedents, Triggers, and Mediators)** of DIGIN dysfunction.
- From there, clinicians take two parallel action steps:
 - **REMOVE Interventions:** Includes symptom relief or referral following clinical guidelines and removal of:
 - **OFFENDING FOODS**
 - **ANTECEDENTS**
 - **TRIGGERS**
 - **MEDIATORS**
 - **REBALANCE Interventions:** Focuses on lifestyle and foundational support including:
 - **SLEEP**
 - **EXERCISE**
 - **CORE FOOD PLAN**
 - **PHYTONUTRIENT SPECTRUM**
 - **STRESS MANAGEMENT**
 - **HEALTHY RELATIONSHIPS**
- If **desired outcomes are not achieved**, the next step is to proceed with **stool testing** for further diagnostic clarity.
- This approach emphasizes addressing likely causes and supporting lifestyle balance before moving to advanced diagnostic testing.

Next Steps

As you continue your training in this course



Gut Dysfunction Stool Testing Treatment Decision Tree



Long Description: Decision Tree

- This decision tree outlines a clinical approach to treating gut dysfunction using stool testing to guide interventions. It is used when signs of maldigestion, malabsorption, gut dysbiosis, or gut inflammation are present.
- **Start** with compelling clinical indicators and A T Ms (Antecedents, Triggers, Mediators) of D I G I N dysfunction.
 - These dysfunctions include **maldigestion, malabsorption, gut dysbiosis, and gut inflammation**.
 - A yellow box notes: "**Stool testing will influence selection of interventions.**"
- Next, begin with:
 - **Symptom relief or referral** as indicated per guidelines
 - **REMOVE interventions** (e.g., removal of offending foods, antecedents, triggers, and mediators).
 - **REBALANCE interventions** (e.g., improving sleep, exercise, Core food plan, phytonutrient spectrum, stress management, and relationships).
- Conduct **stool testing** to inform clinical decisions.
- Based on stool results and/or outcomes from REMOVE AND REBALANCE interventions, apply the **REPLACE, REPOPULATE, and REPAIR** interventions as clinically indicated.
- If **desired outcomes are not achieved** or stool results are **unremarkable**, re-evaluate A T Ms and matrix dysfunctions to refine the treatment approach.
- This graphic emphasizes using stool testing to personalize treatment and the importance of continuous reassessment if the patient does not improve.

FM Approach to DIGIN Dysfunctions: The 5Rs

(addressed in depth in an upcoming session)

- **Remove**
- **Replace**
- **Repopulate**
- **Repair**
- **Rebalance**

Laboratory Evaluation of Gut Dysfunction

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE